

Artificial Intelligence for Depression Assessment: A Comparative Review of Machine Learning and Large Language Models

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Abstract

Depression is a widespread public health issue, and prompt assessment is hampered by subjective self-reporting, restricted access to specialists, and the increasing complexity of multimodal mental health data. In this context, artificial intelligence (AI) has emerged as a promising framework for scalable and data-driven depression evaluation. This study offers a systematic and comparative synthesis of recent works published from 2023 to 2025 about machine learning (ML), deep learning (DL), and large language model (LLM) methodologies for depression analysis. It investigates traditional classifiers such as Support Vector Machines (SVM), Random Forests (RF), AdaBoost, and XGBoost; deep learning architectures such as Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), and Long Short-Term Memory (LSTM); and transformer-based large language models (LLMs) like Bidirectional Encoder Representations from Transformers (BERT) and LLaMA. The review presents a Clinical Readiness and Risk-Aware Comparison Framework (CRRCF) designed to assess methodologies based on task appropriateness, data type, interpretability, and deployment risk, thereby transcending performance-focused reporting. The synthesis indicates that machine learning models are typically more dependable and applicable for organized clinical and psychometric data, while large language models are more proficient in deriving contextual meaning from narrative, conversational, and social media content. Nevertheless, the literature indicates enduring constraints, such as dataset homogeneity, restricted external and prospective validation, bias, privacy issues, inadequate calibration of clinical risk, and the opacity and susceptibility to hallucination in large language models (LLMs). This review provides a clinically informed comparative synthesis and delineates future objectives in multilingual representative datasets, task-specific modeling, multimodal fusion, explainability, and human-supervised deployment for secure, dependable, and contextually relevant depression evaluation.

Keywords: Depression analysis, Artificial Intelligence (AI), Machine Learning (ML), Large Language Models (LLMs), Mental health detection.

1. Introduction

1.1. Background

Depression is a prevalent mental health condition that impacts individuals throughout the world. Initially, depression shows itself as a loss of interest, and then as a long-term sadness along with a multitude of emotional and physical difficulties that establish dysfunction in an individual's life and take away from the quality of life. The WHO [1] states that 1 in 8 individuals in the world are living with some mental disorder, and an extensive number of individuals suffer from some mental disorder because they do not have access to effective treatment. Early detection will decrease the number of suicides and self-harm incidents since this happens more often among those living with depression. Traditional depression prediction

and diagnosis are hampered by a lack of precise biomarkers, reliance on psychological questionnaires and self-reports, which may introduce biases, limited medical resources, financial constraints, and social stigma, which prevent many people from seeking help on time. Academic burnout is another growing issue in higher education, especially among undergraduate students, where there is a strong expectation for academic accomplishment, and performance pressure is high while institutional support is weak. Globally, depression affects more than 300 million individuals [2] worldwide, more than 4% of the population, and is a public health concern. If professionally identified in a timely fashion and treated, depression may lessen the sociological and individual impact of prolonged unrecognised depression. This is especially concerning in environments where there is little help for students or help for students understanding mental health challenges. Bangladesh is an example of a country with intense academic competition and a weak mental health apparatus. In Bangladesh, private students face overlapping stressors [3], both academic, financial, and social, that put them at risk for issues that may escalate to clinical depres-

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sion. Researchers frequently measure depression using the Patient Health Questionnaire-9 (PHQ-9) [4] and the Depression Anxiety and Stress Scales (DASS-21) [5]. Due to advancements in ML, new algorithmic approaches for predicting depression will likely be developed. These approaches are more computationally efficient at extracting meaningful information than typical self-report assessments and provide an objective method of assessing depressive symptom severity compared to traditional methods. However, for the analysis of time series measurements, such analyses typically use classical statistical analysis methods such as correlations and regression, which can miss the complexity and nonlinearities of phenomena related to mental health outcomes. For example, the PHQ-9 is a simple, self-reported questionnaire that can be used to screen for depression, expedite the assessment of depression, and the use of the PHQ-9 allows for monitoring depression progression using ML [6], and LLM are other means of examining high-dimensional data and recognizing hidden patterns. Along with ML-based techniques, LLMs have lately received attention for their ability to interpret high-dimensional and unstructured textual data, making them useful for assessing language cues and complicated mental health indicators. Also, the SVM [7], RF [8], and other ensemble methods have been successful in predicting mental health.

1.2. Motivation

Depression and academic burnout are becoming more widespread mental health issues for college students, especially in low-resource educational contexts [9]. Excessive academic demands, financial strains [10], and lack of positive emotional support [6] can lead to chronic fatigue [11] and disengagement from academics that could progress to clinical depression if not addressed. Although awareness of depression and academic burnout has increased, conventional diagnostic techniques, such as Linear Discriminant Analysis (LDA) [12], are still typically based on linear statistical models of analysis. Consequently, they often fail to account for the complex, nonlinear relationships that exist between burnout and depressive symptoms that students may experience, particularly in low-resource contexts such as Bangladesh. In such contexts, the availability of institutional mental health services for college students is exceptionally limited. The Hamilton Depression Rating Scale (HDRS) [13] offers a valid clinical assessment of depression but is somewhat slow when rated by hand. So, utilizing an ML, DL, LLM approach to using HDRS data can not only reduce the rate of rating, but also provide for better assessment and treatments.

Recently, ML algorithms have shown significant promise in this arena, particularly for analyzing high-dimensional psychometric data. There is much evidence of the predictive efficacy of traditional algorithms, such as RF, Decision Trees(DT), Logistic Regression(LR), eXtreme Gradient Boosting(XGBoost), K-Nearest Neighbors(KNN), and SVM. Still, there is a lack of exploration of the interplay between burnout and depression in the existing research, which limits its applicability. More importantly, there has been less systematic comparison of the two main streams of computational mental health research, which

have progressed in opposite directions: conventional ML techniques and the more recent LLM applications. Because ML and LLM are fundamentally different in their architectural suitability for dealing with organized psychometric measurements and unstructured data acquired from language, voice, and behavioral signals, this discrepancy creates a substantial research gap. To address this issue, this review conducts a comparative assessment of the ML and LLM methodologies for depression analysis. We investigate the various strengths, limits, and potential complementarities of these paradigms by synthesizing information from diverse data modalities such as standardized questionnaires, clinical assessments, natural language, and social media data. The review endeavors to produce empirical findings that may assist in the development of mental health interventions that are more efficient, context-sensitive, and scalable.

1.3. Research questions

The purpose of this systematic literature review was to find out about all the different types of risk-aware AI applications that can be used to test for depression. Specifically, the study looked at ML models, LLMs, and how they handle clinical risk. These questions seek to identify not only present algorithmic approaches, but also the methodological, ethical, and infrastructure obstacles connected with integrating AI in mental health settings.

- **RQ1:** What machine learning and large language model techniques are currently applied in depression assessment, and how do they differ in capability, scope, and clinical relevance?
- **RQ2:** How do methodological, ethical, and infrastructural challenges influence the safe implementation of AI-based depression assessment in real-world settings?
- **RQ3:** What methodological mechanisms enable AI-based depression assessment systems to incorporate risk-awareness for reducing misdiagnosis and promoting patient safety?
- **RQ4:** Which strategies are most effective for facilitating the clinical adoption of risk-aware AI systems for depression assessment, and what trade-offs do they involve?

1.4. Criteria for inclusion and exclusion

In order to restrict the analysis to esteemed, high-quality studies, we applied a systematic literature review (SLR) framework [14]. The SLR guidelines that we adhered to mean that we only examined studies that utilized ML and LLMs [15] for assessment, monitoring, or detection of depression. Table 1 discusses the inclusion and exclusion criteria for selecting articles.

During the literature review, we exclusively investigated peer-reviewed journal papers from reputable academic databases (e.g., *IEEE Xplore*, *ScienceDirect*, *SpringerLink*, *PubMed*, *Scopus*, and *Google Scholar*). Our selection of articles published between 2023 and 2025 reflects our commitment to offering cutting-edge research in this rapidly evolving field.

These works collectively provide a strong empirical basis for comprehending the strengths and weaknesses of ML and LLM methodologies in depression evaluation. We specify the essential foundation for the comparative, risk-conscious synthesis and analysis in this study, directing the recognition of dominant patterns, methodological deficiencies, and prospective research directions.

2. Literature Review

Table 2 summarises recent findings between 2023 and 2025, highlighting a growing interest in utilising ML approaches to predict mental health issues from survey data, particularly in academic and high-stress environments. Across these studies, psychometric questionnaires remain the dominant data modality, reflecting their accessibility as well as inherent limitations related to self-reporting bias and contextual variability. Schaab et al. [16] conducted an extensive systematic review of major academic databases to evaluate the methodological quality and evidence strength of ML-based mental health studies. Their findings emphasised the importance of transparent reporting, appropriate validation strategies, and bias-aware study design, noting that although many ML models achieve high predictive accuracy, the overall quality of evidence is frequently constrained by limited dataset sizes and a lack of external validation. Focusing on early detection of psychological distress among students, Hermawan et al. [17], and Sherpa et al. [18] employed supervised classifiers such as SVM, RF, and DT using academic survey data. These studies demonstrated encouraging performance in identifying stress, anxiety, and depressive symptoms, supporting the feasibility of ML-assisted screening tools in educational settings. Similarly, Bhatnagar et al. [19] applied multiple conventional ML classifiers to anxiety detection in university students and reported superior performance from RF models, reinforcing their frequent selection in survey-based mental health prediction tasks. The COVID-19 pandemic further accelerated research examining population-level stressors and their psychological impacts. Nooripour et al. [20] applied neural-network-based and conventional ML techniques to classify pandemic-related anxiety, stress, and depression symptoms, achieving moderate predictive performance. Theodoros et al. [21] similarly evaluated ensemble classifiers using k-fold cross-validation, illustrating the adaptability of ML methods to rapidly collected and heterogeneous survey datasets. Beyond predictive performance, several studies explored methodological extensions and contextual considerations. Saleem et al. [22] examined ethical challenges associated with ML-based mental health systems, emphasising fairness, transparency, informed consent, and safeguards against algorithmic bias. Zulfiker et al. [23] proposed a stacked ensemble framework to improve prediction stability, while Al-Mamun et al. [24] demonstrated that incorporating demographic and behavioural features alongside psychometric measures can enhance generalisation performance. More recent work has expanded ML-based mental health analysis beyond traditional tabular survey data. Prelong et al. [25] validated ML models using longitudinal health records, supporting their applicability for longer-term outcome

prediction. Ciharova et al. [26] explored multimodal stress detection using physiological and contextual data, revealing both the potential and practical challenges of continuous monitoring. Additionally, Zhang et al. [27] and Sharma et al. [28] introduced DL and hybrid optimisation-based models, achieving improved predictive accuracy compared to classical classifiers, albeit often at the cost of increased computational complexity and reduced model transparency.

The reviewed literature demonstrates that ML techniques are effective for modelling depression, anxiety, and stress in controlled research settings. However, most studies rely on internally validated survey datasets, provide limited reproducibility analysis, and lack systematic evaluation of bias and deployment feasibility. These limitations underscore the need for structured comparative reviews that assess not only predictive performance, but also data requirements, validation robustness, explainability, and readiness for integration with emerging language-based and foundation model approaches. Overall, the body of work consolidated in Table 2 indicates that reported performance gains are frequently influenced by preprocessing decisions such as feature selection and class balancing, rather than model architecture alone. Moreover, the predominance of internal validation strategies and student-centric datasets suggests that existing results should be interpreted as indicators of methodological feasibility rather than definitive evidence of real-world clinical readiness.

3. Survey Methodology

3.1. Application Area

In the specific area of research on the analysis of depression, it is abundantly clear how helpful computational methods can be in mental health. Both LLMs and ML methods show promise in advancing our understanding of depression; however, they have different strengths and advantages depending on their individual contexts of usage. ML methods have demonstrated considerable success with structured data, for example, from contexts of electronic health records or behavioural monitoring, whereas LLMs represent an advancement in the field because they allow examination of complex unstructured text data at scale. This section highlights each method’s unique contributions to depression research and clinical application based on their respective research applications. Figure 2 provides an overview of the application areas where ML and LLM have been employed for depression analysis.

3.1.1. Clinical Decision Support

Machine learning (ML) is increasingly used in clinical settings with large structured health data [?]. Various algorithms have been developed for the purpose of detecting depression through the analysis of medical records, comorbidities, medications, and self-report surveys. To improve diagnostic and prognostic accuracy, ML uses hidden patterns in numerical, categorical, or binary data. For example, classifiers trained on electronic health records can detect high-risk patients who might

Table 1: Inclusion and exclusion criteria used for selecting articles in this review.

Category	Inclusion Criteria	Exclusion Criteria
Types of study	Original research and review papers focusing on depression detection, monitoring, or analysis using ML or LLM approaches.	Opinion pieces, editorials, theses, white papers, and non-peer-reviewed sources.
Language	Articles published in English.	Duplicate or non-English studies.
Publication year	Studies published between 2023–2025, to ensure coverage of recent developments.	Articles outside this time frame or unrelated to the review focus.
Source	Peer-reviewed journals and conference proceedings.	Papers lacking sufficient methodological detail or not meeting review standards.
Methods	Studies employing ML, DL, or LLM techniques for depression analysis.	Studies using only traditional statistical methods without ML or LLM.
Region	No restriction to specific geographic locations.	–
Settings	Clinical records, social media texts, conversational datasets, or multimodal mental health data.	Studies based solely on non-textual data (e.g., EEG, neuroimaging) without ML or LLM integration.

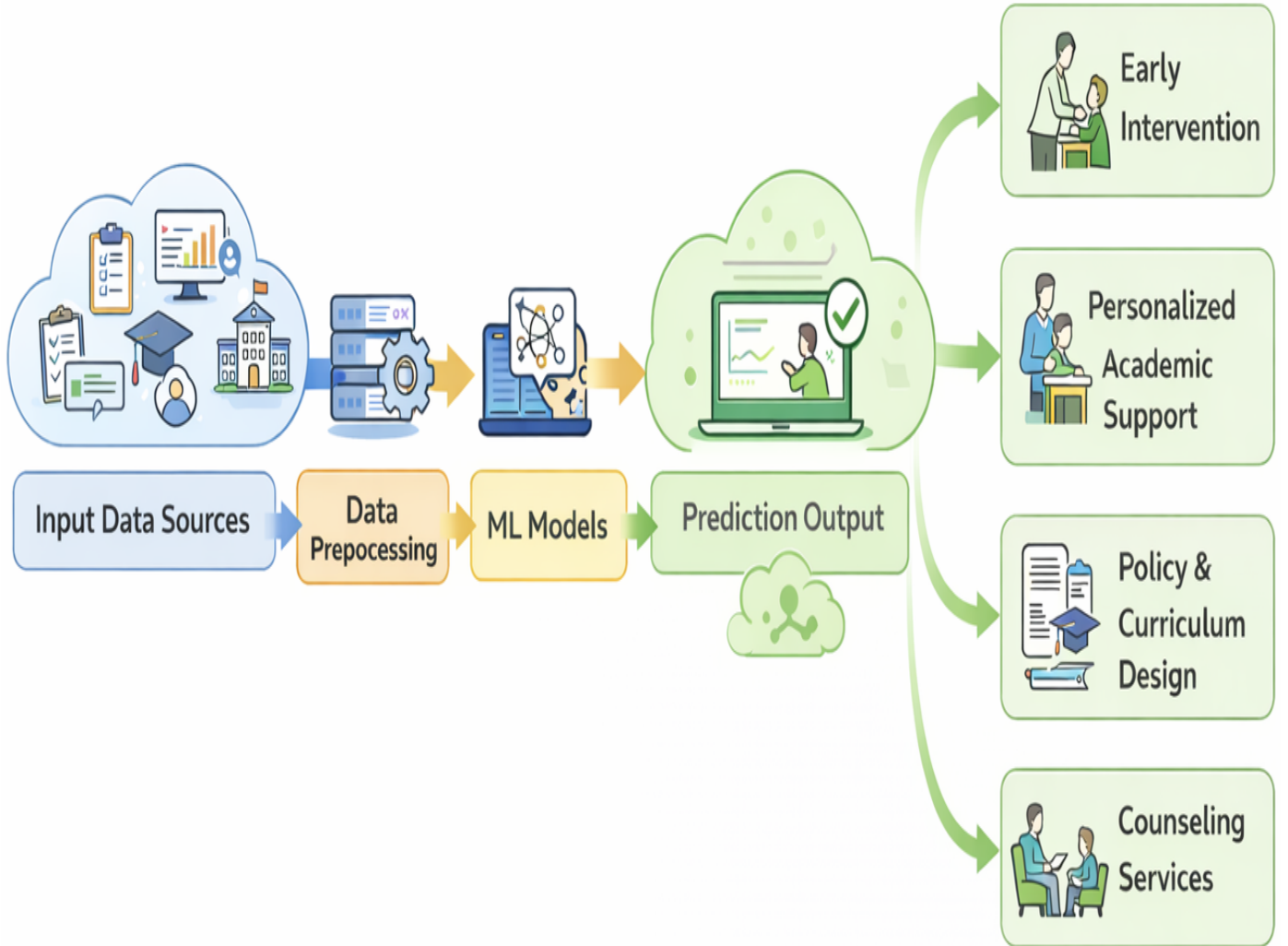


Figure 1: Conceptual workflow of an ML-based depression prediction framework.

Table 2: Comparative performance and limitations of ML-based depression and mental health survey studies from 2023 to 2025.

References	Datasets	Conventional Approaches	ML-based Techniques	Results	Methods	Contribution
Schaab et al. [16]	✗	✗	✗	✓	Conduct a PRISMA-based systematic review, with a protocol registered in PROSPERO, and search major databases such as PubMed, Embase, Web of Science, PsycINFO, and assess quality of the evidence using GRADE, along with assessing both the performance of ML as well as the quality of studies.	Synthesise the current science on using ML models to detect mental health problems in undergraduate students. Specifically, they conclude that while ML models have been shown to display high accuracy in detecting undergraduate students' mental health needs, the quality of evidence is typically low, and studies rarely incorporate an external validation.
Hermawan et al. [17]	✓	✗	✓	✓	DASS-42 survey data from the completion of 396 students was used where, after the requisite preprocessing of one-hot encoding, normalizing data, and balancing the dataset with SMOTE, to use with SVM and RF models that we trained and tested.	They introduced a lightweight framework based on SVM with SMOTE, which highlights minority class detection and achieves high classification accuracy for the early detection of depression, anxiety, and stress among university students.
Sherpa et al. [18]	✓	✗	✓	✓	Used survey data from university students, applied preprocessing and feature selection, implemented multiple ML models (DT, RF, SVM, LR, NB), evaluated using accuracy and other metrics.	Identified key depressive risk factors among students, compared multiple ML models, and highlighted the most important predictors for early intervention strategies.
Nooripour et al. [20]	✓	✗	✓	✓	Collected survey data on COVID-19 stress and demographics, then used ANFIS and eight ML algorithms to classify stress levels, with DT performing best.	Applied ML to assess COVID-19 stress in Iranian students, identifying key demographics and demonstrating ML's potential for early detection and targeted support.
Saleem et al. [22]	✗	✓	✗	✗	Literature review and ethical analysis of ML used for depression and anxiety, discussed privacy, consent, bias, and transparency issues.	Provided a framework of ethical guidelines and considerations for deploying ML-based mental health systems.
Zulfiker et al. [23]	✓	✗	✓	✓	Collected survey data from 604 participants; applied feature selection (SelectKBest, mRMR, Boruta) and SMOTE to balance classes; trained six ML models—KNN, AdaBoost, Gradient Boosting, XGBoost, Bagging, and Weighted Voting.	Proposed a model-mediated stacked ensemble method that improved prediction performance for mental health classification compared to individual models. AdaBoost with SelectKBest achieved 92.56% accuracy, demonstrating that feature selection and data balancing significantly improve ML-based depression prediction.
Al-Mamun et al. [24]	✓	✗	✓	✓	Collected survey data on sleep duration, insomnia, and geographical details from prospective university students, preprocessed and analysed data, applied DT and RF, evaluated performance using accuracy and related metrics.	Showed how ML models can be used with geographical and behavioural data to identify students at risk of insomnia and guide preventive health measures.
Prelong et al. [25]	✓	✗	✓	✓	Used electronic health records from emergency department patients with chest pain, performed preprocessing and feature selection, trained and validated multiple ML models, evaluated using AUROC and related metrics.	Validated an ML model to predict 30-day adverse outcomes for chest pain patients, showing high predictive accuracy and clinical applicability.
Ciharova et al. [26]	✓	✗	✓	✓	Administered an online Trier Social Stress Test to 42 students, capturing visual, acoustic, verbal, and physiological data, extracted and reduced to 30 relevant features, predicted stress severity using Bayesian Ridge regression and LR, validated with leave-one-out cross-validation.	Introduced an online stress induction protocol with pre-recorded interviewers, demonstrating the potential of multimodal data for continuous stress detection and outlining methodological challenges for future work.
Theodoros et al. [21]	✓	✗	✓	✓	Collected DASS-21 responses from students during the COVID-19 lockdown in Greece, applied multiple ML classifiers (IBk, RF, MLP, FURIA, SMO), and evaluated with 10-fold cross-validation.	Showed that ML classifiers, particularly RF and FURIA, can accurately detect mental health conditions from questionnaire data, supporting early identification and intervention.

References	Datasets	Conventional Approaches	ML-based Techniques	Results	Methods	Contribution
Zhang et al. [27]	✓	✗	✓	✓	Collected student mental health questionnaire data and used supervised ML models to predict risk levels, validated with accuracy and performance metrics.	Delivered an AI tool for assessing mental health risk among students, enabling timely support and preventive action.
Sharma et al. [28]	✓	✗	✓	✓	Used preprocessing and feature selection on psychometric data, then trained a hybrid AOA-CNN-XGBoost model, outperforming CNN and AOA alone.	Introduced a hybrid DL and optimisation-based model for predicting student stress, improving accuracy over existing methods.
Bhatnagar et al. [19]	✓	✗	✓	✓	Collected anxiety-related data from university students using Likert scale questionnaires, applied NB, DT, RF, and SVM to classify anxiety levels, compared predictive performance.	Demonstrated that ML can effectively classify anxiety levels among students, with RF achieving the highest accuracy and providing a basis for targeted mental health interventions.

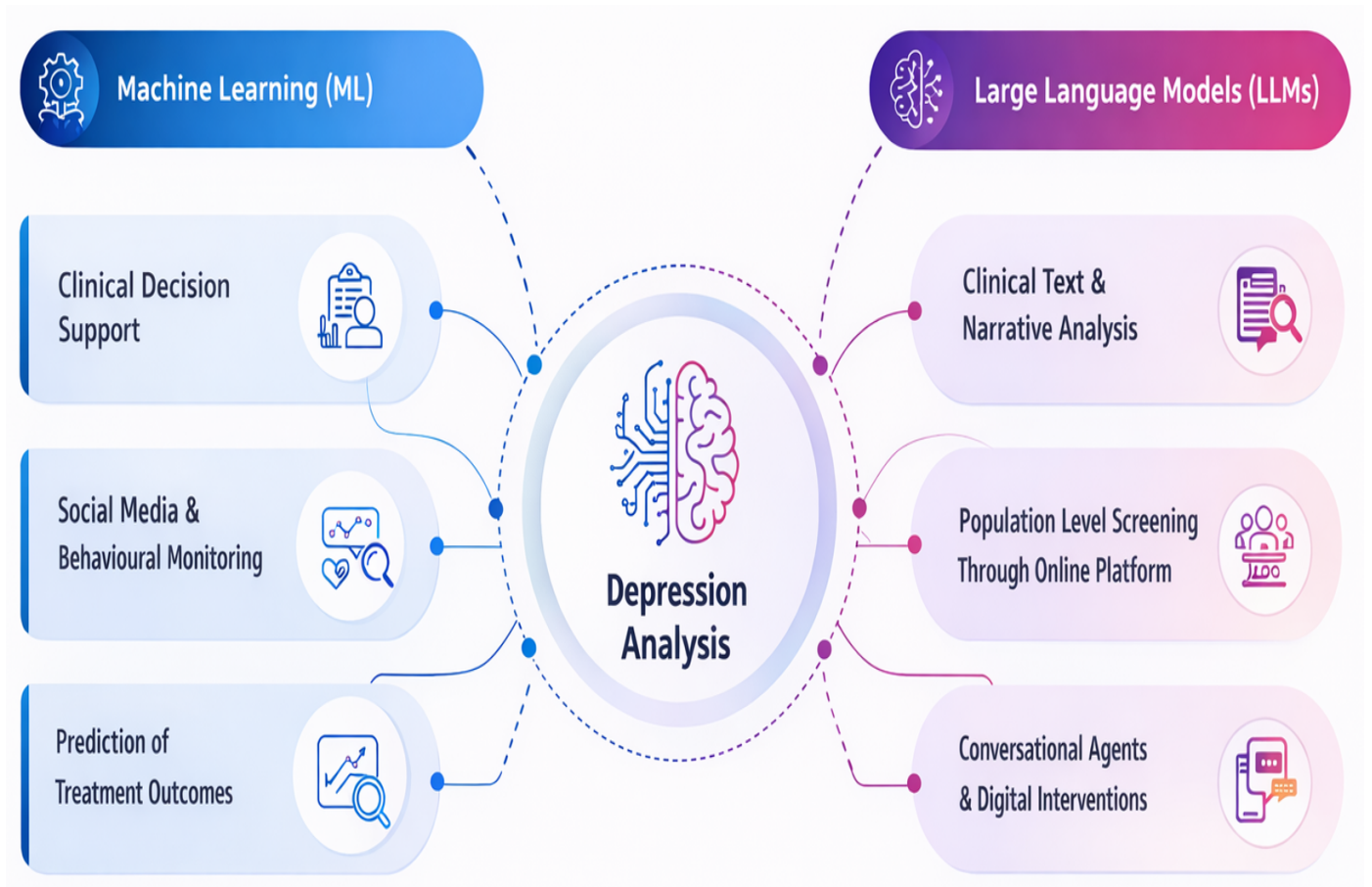


Figure 2: Exploring the Applications of ML and LLM in Depression Detection and Analysis.

otherwise go unnoticed during routine appointments. In addition, ML allows for the early diagnosis of depression based on social media behavior. Research has demonstrated that linguistic patterns in postings [29] and engagement frequency [30] can consistently indicate risk, giving clinicians valuable information for prompt intervention.

In conventional healthcare, where mental health screening is limited due to consultation time constraints, such approaches are especially beneficial. ML can lessen the uncertainty of subjective diagnoses and make assessments more consistent. Nevertheless, there are still obstacles to overcome, such as the fact that clinical datasets are prone to bias and often have limited applicability across different populations. The conceptual process of a system for depression prediction using ML is shown in Figure 1.

3.1.2. Social Media and Behavioural Monitoring

The second notable use of ML is in the analysis of digital footprints, specifically in social media [31]. ML can find evidence of early markers of depression using linguistic features, posting frequency, and interaction tendencies. Most of the previous studies in this area identified and quantified evidence of distress using engineered features, like total negative words, first-person pronouns, or amount of engagement, and then modeled them using classifiers such as LR or RF. In conclusion, while somewhat limited in that they lost nuance and contextual meaning, they did help pioneer applying mental health monitoring at population scales.

These tools are useful for pinpointing specific at-risk groups, tracking the pathway of psychosocial distress over a collective population, such as how far psychosocial distress extends during a crisis, and creating data for early intervention. As our society continues its gradual transition towards digital interactions as part of everyday living, utilizing ML to perform behavioral surveillance has become necessary due to the failure of traditional psychological surveys in determining the trend of population health and well-being within defined populations. Additionally, as the method of using ML technology for this purpose will likely increase in prevalence and importance, there are also ethical issues that must be addressed regarding privacy rights for data collection, informed consent from users, and the appropriate use of collected information.

3.1.3. Prediction of Treatment Outcomes

Machine learning has been progressively utilized to forecast therapy outcomes in depression [32]. It can forecast the likelihood of relapse, classify patients into subgroups with shared patterns of recovery, and improve clinical documentation by means of text mining when fed longitudinal data such as therapy attendance, medication adherence, and symptom trajectories. These findings pave the way for adaptive care, which in turn facilitates better clinical decision-making and treatment programs that are constantly customized.

Additionally, predictive models can help improve treatment precision by informing patients about the relative benefits of medication and psychotherapy. The broad use of ML-driven

insights is still hindered, however, by the persistent issues of doctor trust and patient acceptance.

3.1.4. Clinical Text and Narrative Analysis

The advancement of LLM has greatly influenced the evolution of unstructured data, especially in the context of mental health. A patient's intricate moods and mental states can be expressed in various ways: their clinical notes, therapy session transcripts, and even self-reports may contain such information. However, capturing this information is not possible with traditional ML. Models like BERT and RoBERTa, their generative relatives, and the appropriate advanced models are maintained and improved upon to help grasp such information and allow healthcare to understand patients better. So, the LLM applications increasingly focus on supporting clinical treatments [33] and interventions, positioning them as complementary tools in modern mental healthcare. With the help of LLMs, symptom severity can be evaluated through narrative medical records, cognitive distortions can be detected, and therapy sessions can be meaningfully condensed. Doing all of this brings a whole new set of instruments to the providers, but integrating them within clinical care must address issues of trust, safety, and adherence to privacy laws. Due to the continued improvements in contextual understanding, LLMs may soon be real-time diagnostic assistants, which, during patient consultations, can highlight issues that require further evaluation.

3.1.5. Population Level Screening Through Online Platforms

LLMs have demonstrated significant ability in identifying signs of depression on social media and other online platforms [34]. Unlike the older ML methods that relied on weighted text and user interaction features, LLM can analyze posts and comments with greater understanding by incorporating semantic and contextual analysis. The systems can analyze large chunks of text and identify signs of emotional instability, such as hopelessness and emotional withdrawal, which would be harder to identify with shallow models. This has helped a lot in tracking the mental state of the younger generation, who tend to be more freely expressive on social media in comparison to structured offline interactions. In addition, LLM-based systems facilitate instantaneous monitoring of an entire population. Consequently, public health officials can be provided with the necessary tools to identify new developments in the psychological state of populations and respond promptly. For instance, during the COVID-19 pandemic, attempts were made to utilize these systems to monitor changes in the mental health of the population. Nonetheless, the concerns of consent, privacy, and the possibility of abuse must be dealt with in advance, before the widespread implementation of these systems can be allowed.

3.1.6. Conversational Agents and Digital Interventions

The integration of LLMs would be an amazing advantage for digital therapeutic services due to their innovative ways of providing digital healthcare assistance to users. An example of the type of assistance LLM-driven chatbots could provide is through chatting in an informal manner with users, such as

Psycho-Education, providing emotional support, and delivering different types of therapy, including Cognitive-Behavioural Therapy. Additionally, because LLM-powered chatbots are continuously available, this makes them the logical access point for addressing the many barriers to accessing mental health care based on limited provider availability, distance from provider office locations, and stigma against mental illness. In contemporary research, it has been demonstrated that these agents can be tailored to individual users through the use of conversational analytics along with behavioural data from smartphones or wearables. LLM-based agents [35] can offer empathetic interactions and modify offered treatments in an adaptive fashion, thus enhancing existing therapy and serving therapy-devoid populations.

However, ensuring their safe use entails clinical trials, achieving regulatory standards, and safeguards against harmful or misleading speech. When integrated in an ethical framework with mental health practitioners, these tools can be used as adjuncts that handle some aspects of care, thus broadening and simplifying access as well as improving continuity of care.

4. Datasets and pre-processing

Datasets and pre-processing form the backbone of any computational approach to depression analysis. In this section, we first discuss the datasets most frequently used to train and evaluate ML and LLM systems for depression detection. Following this, we examine the pre-processing strategies that transform raw data, whether clinical records, text transcripts, speech signals, or social media content, into a usable form for computational modeling.

4.1. Datasets

For research on depression identification and analysis to advance, trustworthy datasets must be available. These databases serve as standardized tools that let academics confirm results from different investigations and compare approaches. Datasets used in depression analysis might range from clinical records and psychiatric evaluations to multimodal resources such as social media, speech, or text data. Benchmark datasets ensure that performance improvements can be objectively evaluated by giving ML and LLM-based models a common foundation for evaluation. Table 3 provides an overview of the most commonly used datasets in this field.

4.2. Pre-processing Techniques

To ensure that computational models can accurately analyze raw data, pre-processing is an essential step. Pre-processing for ML methodologies may involve signal denoising, dimensionality reduction, feature scaling, or missing-value imputation. In contrast, typical pre-processing for LLMs includes tokenisation, stop-word removal, lemmatisation, and normalizing terms, for example, converting social media slang to standard terminology. For multimodal data, additional modality-specific pre-processing steps are often required, such as text scrubbing,

audio feature extraction, or coding of clinical data. These procedures directly influence the fairness, generalisability, and accuracy of depression analysis models. A concise overview of the pre-processing methods employed in ML and LLM studies is provided in Table 4.

4.3. Dataset Bias and Generalisability Challenges

Depression research has used a wide range of data sources, although differences in experimental design and data gathering limit the generalizability of many datasets. Most research uses self-report measures like the PHQ-9, which have limitations in terms of the reliability of ground-truth labels utilized to train models due to issues including recall bias, underreporting, and social desirability effects. Similarly, datasets derived from social media mainly consist of English-speaking, well-off people and are culturally and linguistically homogeneous. This limits the capacity of models to identify psychological distress in diverse populations and makes it impossible to capture culturally specific stigma associated with mental health. As a result, models that are trained on these datasets would not give fair evaluations in low- and middle-income nations like Bangladesh, where healthcare systems and cultural values are different. To overcome these restrictions and guarantee accurate and fair assessment on a worldwide scale, it is necessary to create datasets in several languages, validate them across cultures, and test depression detection methods in low-resource environments.

5. Algorithms for Depression Analysis

In depression analysis, algorithms are needed to pull out features, make predictive models, and figure out what semantic trends mean. Most of the time, structured data sources like social media posts, poll results, and medical records are used by traditional ML methods. In contrast, LLMs make use of unstructured text (US) for contextual embedding and semantic embedding. A review of several types of algorithmic approaches was performed to highlight the distinct methods, advantages, and disadvantages of each approach when working with data from multi-modal sources related to depression.

Figure 3 illustrates a generalized ML-based depression analysis framework, highlighting the interaction between feature extraction, learning algorithms, and predictive outputs.

5.1. Machine Learning Algorithms

The application of ML techniques continues to be widespread for the identification of depression, particularly in cases where datasets are limited and features are engineered. Popular ML algorithms include:

5.1.1. Support Vector Machines (SVM)

Support vector machines [48] are widely used supervised ML models in depression research, particularly for classification tasks involving structured and high-dimensional feature sets. It constructs an optimal decision boundary by maximizing the margin between classes, in a high-dimensional feature space [49]. In depression analysis, SVM has been applied to diverse

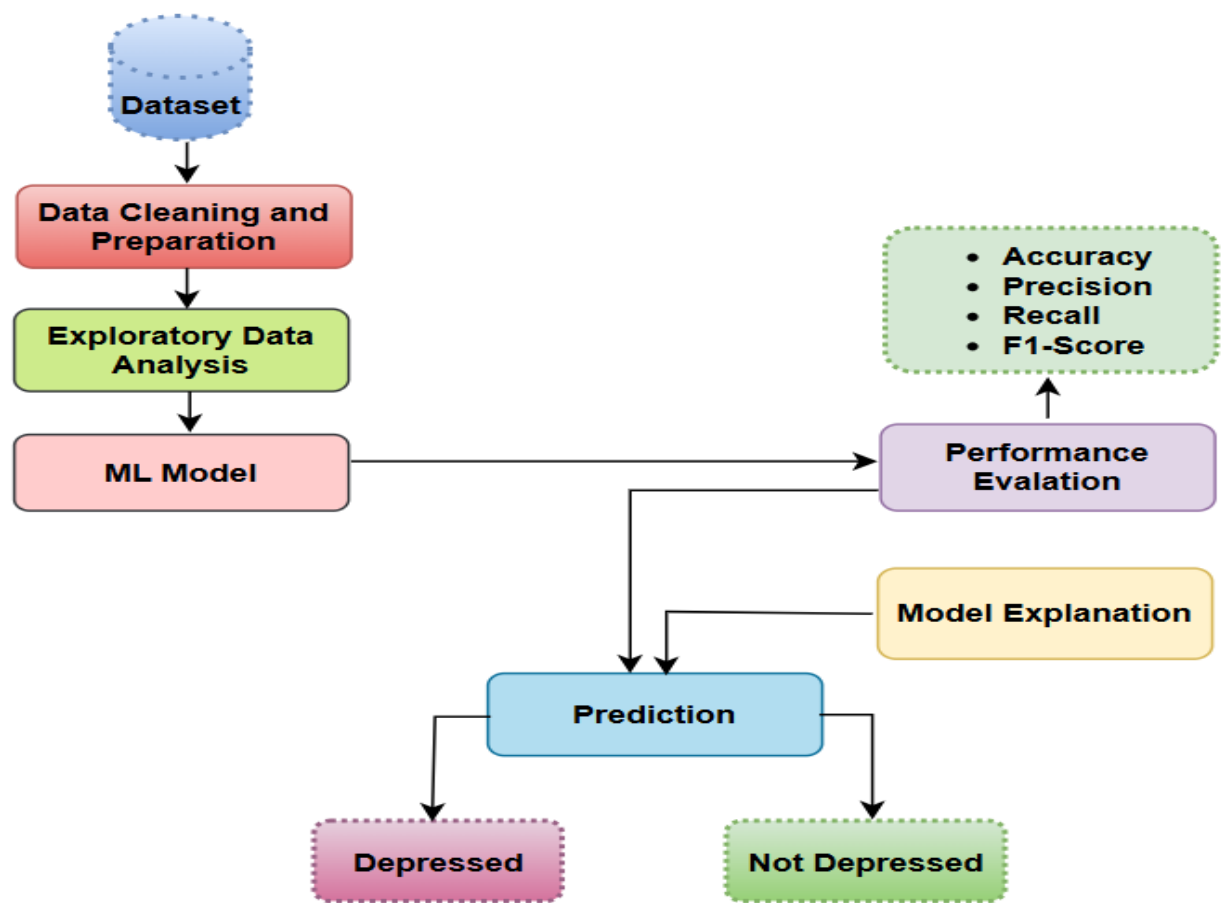


Figure 3: Generalized ML framework for depression prediction.

Table 3: ML and LLM-based Datasets for Depression Analysis

Dataset	Type	Domain	Details	Reference
PHQ-9	Questionnaire	Academic	Based on Patient Health Questionnaire (PHQ-9) responses from students, commonly used for depression severity classification.	[36]
NHANES 2013–2014	Survey	Healthcare	National Health and Nutrition Examination Survey with PHQ-9 depression scores and health attributes.	[37]
DAIC-WOZ	Multimodal (Audio, Video, Text)	Clinical Interviews	Collected during psychological interviews. Widely used for AVEC depression challenges.	[38] [39]
AVEC	Multimodal (Audio, Video, Physiological)	Clinical or Challenge Dataset	Benchmark dataset from Audio or Visual Emotion Challenge, includes depression severity labels.	[40]
eRisk	Text (Forum Posts)	Online Forums (CLEF)	Early risk detection dataset from CLEF shared tasks. Focused on depression and suicidal ideation.	[41]
RSDD	Text (Reddit posts)	Social Media	Posts from Reddit users with self reported depression diagnosis labels.	[42]
Twitter Depression Dataset	Text (tweets)	Social media (Twitter)	Contains tweets labeled for depressive symptoms, suicide ideation, and emotional states. Useful for fine-tuning LLMs in short text classification.	[43]
CLPsych Shared Task Data	Text (forum posts)	Computational Linguistics Clinical Psychology workshops	Annotated datasets from Reddit, Twitter, and online forums used in shared tasks for detecting mental health conditions. It includes depression and PTSD.	[44]
SMHD	Text	Mental health self-disclosure	Large dataset of Reddit users who explicitly disclose a mental health diagnosis (including depression, anxiety, bipolar disorder). It enables LLMs to learn linguistic cues from authentic disclosures.	[45]
MD-Dialogue or Counseling Conversations	Text	Therapy and counseling datasets	Contains conversational exchanges between patients and counselors. Useful for training conversational LLMs to provide empathetic, context-aware responses.	[46]
Synthetic Augmented Datasets	Text (LLM generated)	Augmented clinical and social media	Datasets generated using GPT-based augmentation to balance class distribution in depression detection tasks. Often combined with real corpora to improve performance.	[47]

data sources, including physiological signals, behavioral features from wearable devices, and linguistic attributes extracted from social media and self-reported assessments such as PHQ-9. Their ability to model non-linear relationships through kernel functions makes them suitable for heterogeneous mental health datasets.

One of the biggest advantages of using SVM is its excellent performance and accuracy when working with small-to-medium-sized datasets. This is particularly beneficial for mental health research because of the often limited data availability. While SVMs provide excellent results, they often have difficulties when dealing with hyperparameter tuning, particularly as the datasets grow larger. Consequently, SVMs are frequently combined with other techniques, such as feature selection and dimensionality reduction, to create more robust systems, and hybrid methods are becoming more common as they enhance the practical implementation of SVMs in depression screening applications [50]. Figure 4 shows the general layout of the SVM-based mental health detection system, outlining the primary steps such as data gathering, preprocessing, training the model, and lastly, evaluation of the model for mental health prediction.

5.1.2. Random Forests (RF)

Random Forest [8] are widely used ensemble learning model in depression research due to their robustness against overfitting and ability to capture non-linear relationships in heterogeneous datasets. Their suitability for integrating clinical, sociodemographic, and biological features, along with the provision of feature importance measures, has contributed to their frequent adoption in depression-related studies.

Empirical evidence indicates that RF performs effectively in predicting depression severity and risk across diverse settings. To mitigate class imbalance, weighted RF variants and resampling techniques such as the Synthetic Minority Over-sampling Technique (SMOTE) [51] have been applied, leading to improved area under the curve (AUC) performance [52]. RF-based models have also been employed in electronic health record analyses, where they reveal non-linear associations between intervention patterns and symptom outcomes [53].

Despite their strengths, RF faces limitations in scalability for very large datasets and may offer reduced transparency compared to simpler linear models such as LR. Comparative studies suggest that while RF achieve competitive performance, gradient boosting approaches may outperform them for specific prediction tasks. Overall, RF remains well-suited for structured, tabular depression data, whereas LLMs are better aligned with unstructured text-based analysis. Figure 5 illustrates an overview of the automated depression analysis system that makes use of the RF methodology to improve the accuracy of diagnostics.

5.1.3. Gradient Boosting

Gradient Boosted Machines [54] are a form of ML technique that sequentially pays attention to those incorrectly classified data points in the previous round to improve their predictive performance in later rounds. A major strength of GBM

compared to other standard ML techniques is that it has high predictive accuracies and, in the majority of research studies, the predictive accuracies of GBM exceeds that of other traditional ML techniques. As a result, Gradient Boosting is an ideal approach for researchers who are looking into studying depression through Large, complex data sets such as Bio Banks or Electronic Health Records (EHR).

Overall, gradient boosting is particularly effective in depression analysis scenarios where maximising predictive performance is a primary objective, especially for the detection of depression from social media posts [55]. While LR may serve as a more stable and interpretable baseline, GBMs offer a performance-oriented alternative for developing accurate clinical decision-support models from structured data. In order to analyze patterns connected to mental health, a thorough ML framework is shown in Figure 6.

5.1.4. Logistic Regression (LR)

The Logistic Regression method and its crucial significance in the study of clinical depression [56] are thoroughly examined in this review. The probability of a categorical outcome, such as a clinical diagnosis of depression, based on a set of independent variables can be effectively predicted using LR, a foundational statistical model commonly used for binary classification tasks. It is a critical component in developing clear and practical predictive models for mental health applications due to its ease of use, computational efficacy, and, most importantly, great interpretability. The strengths of LR are highlighted, particularly its capacity to provide direct insight into the relationship between clinical and demographic factors and the risk of depression. The model's coefficients allow clinicians and researchers to easily understand the influence of each predictor, a significant advantage in clinical decision support. However, key limitations are also addressed, including the assumption of a linear relationship [57] and its potential underperformance compared to more complex, non-linear models on high-dimensional data.

Finally, findings from a variety of research are summarized to assess LR prediction performance on diverse depression datasets. Optimal practices for model development are discussed, such as effective feature selection, regularization approaches, and rigorous evaluation using measures like AUC-ROC. While more complex algorithms may provide minor performance improvements, LR's unequalled interpretability and durability ensure its continued importance as a critical baseline model and a potent instrument for generating therapeutically significant insights [58]. Figure 6 highlights the comparative performance of several ML models for detecting depression based on social media posts and compares the performance of several traditional ML approaches.

5.1.5. K-Nearest Neighbors(k-NN)

This review assesses the k-Nearest Neighbours (k-NN) algorithm [59] as a tool for research regarding clinical depression. Instance-based learning is what k-NN does. It creates depressive diagnoses using non-parametric techniques with both Clinical and Psychometric Data Sets. Because the k-NN Techniques which include interpretability and accuracy across small data

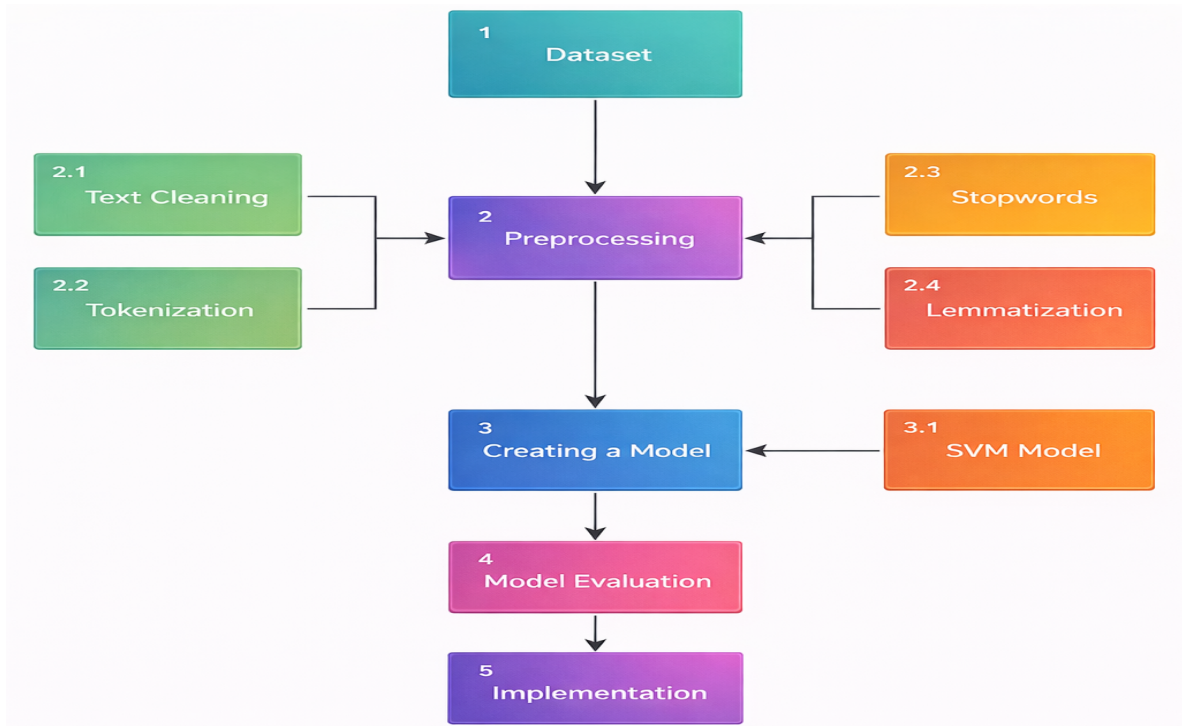


Figure 4: SVM-based mental health detection [50].

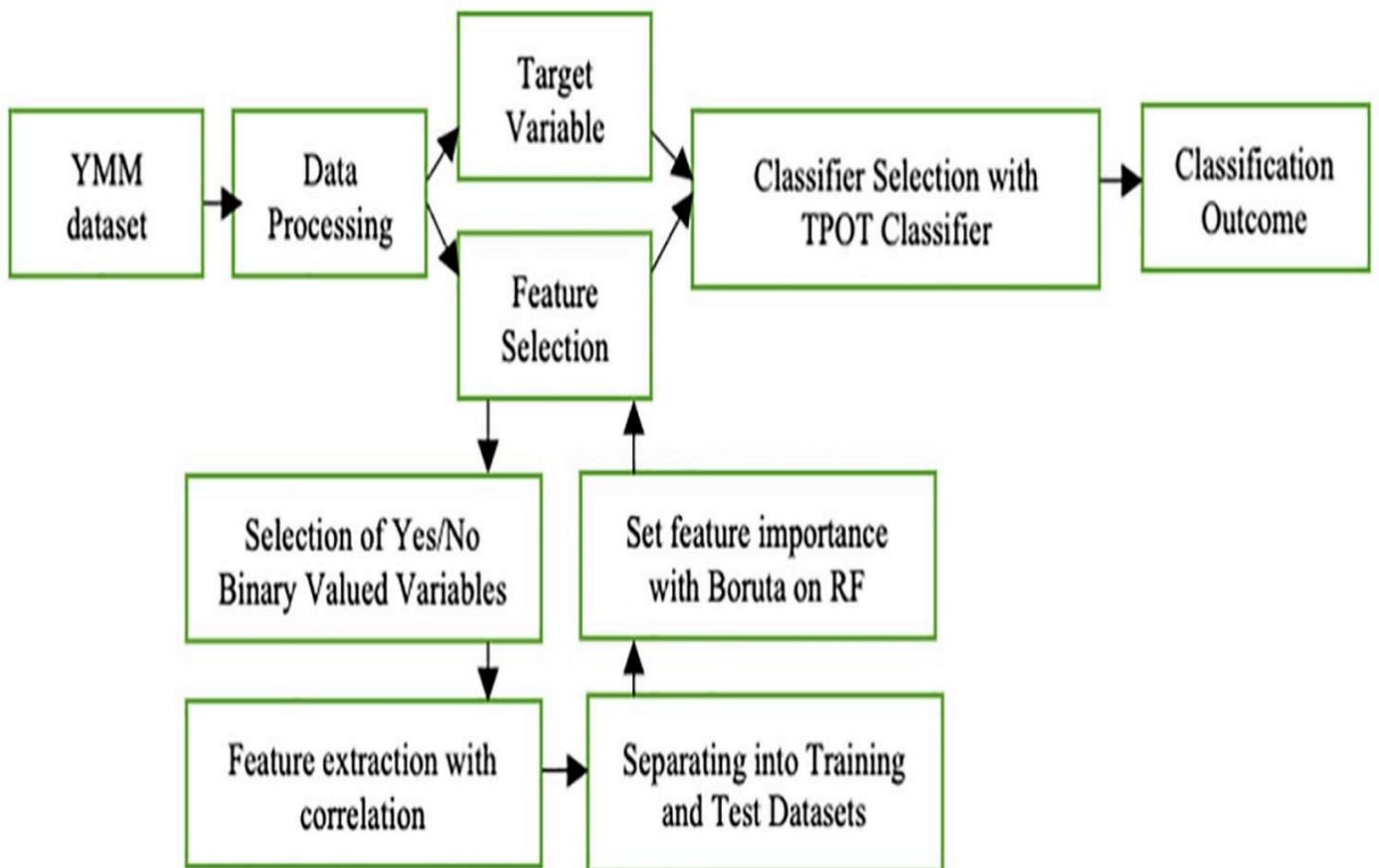


Figure 5: Functional workflow of the depression detection method [53].

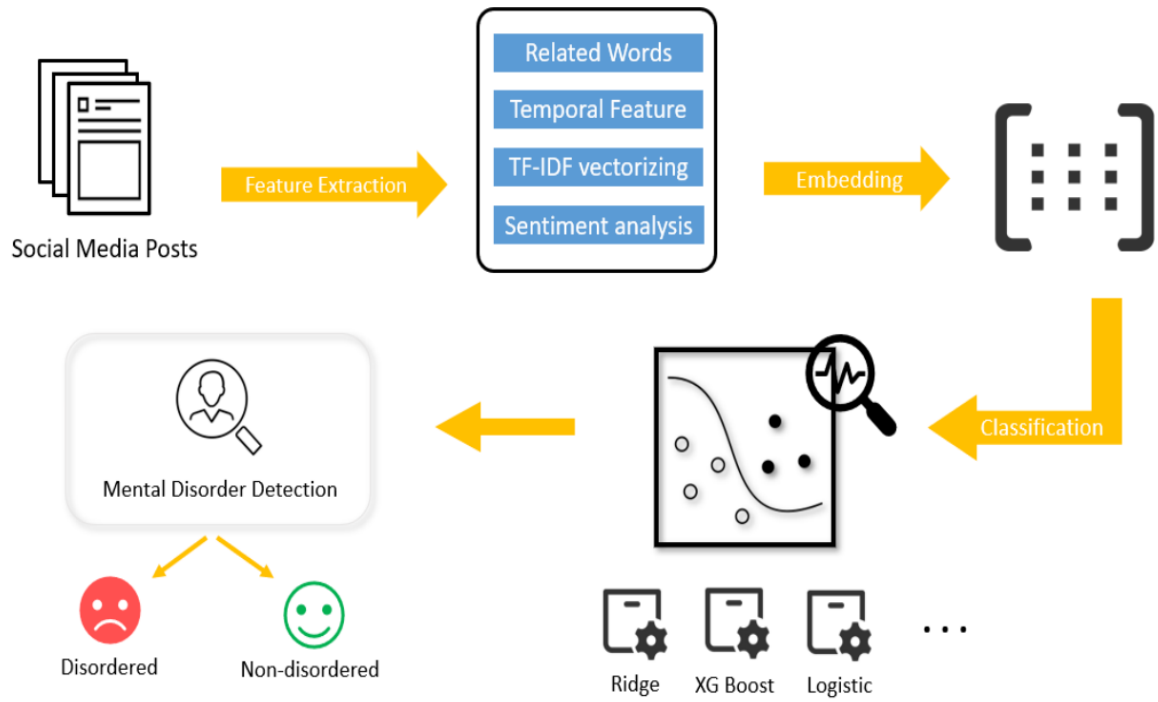


Figure 6: Efficient machine learning framework for analyzing mental health patterns [55].

sets, have an established track record in doprium research, a large number of researchers have conducted or continue to conduct research using k-NN techniques in various stages, such as clinical data sets and psychometric data sets for development. Many comparative studies show that k-NN yields reasonable levels of predictability when applied to smaller data sets, yet it is consistently shown to provide lower levels of ultimate predictive power than ensemble methods such as RF and Gradient Boosting when applied to larger, richer datasets that contain many more features.

Some potential areas for future improvement include combining k-NN with other methods or more sophisticated techniques [60] for measuring distance, to improve k-NN's effectiveness as a clinical decision-support tool for mental health disorders like depression. Figure 7 depicts a system that may automatically detect depression and make use of ML to aid in early diagnosis and treatment planning. This system is connected with mental health applications.

5.2. Deep Learning Algorithms

Deep learning [61] is a potential model for mental health analytics like predicting depressive symptoms, but its main benefit is creating a full end-to-end model of raw, unstructured, multi-dimensional data types, including audio, video, and text data. This is a big improvement over standard ML techniques, which frequently rely on human-picked characteristics. DL models automatically learn hierarchical and complicated representations, capturing nuanced patterns and temporal correlations that manual engineering cannot detect.

Concatenating embeddings horizontally enables the integration of multimodal information. Combined audio and text em-

beddings are used to distinguish depressive patients from healthy individuals by capturing both short-term and long-term linguistic and auditory characteristics. Finally, the concatenated embeddings produced by the 1D CNN and BiLSTM models are fed into two single-layer fully connected networks: one estimates the severity of depression, while the other outputs a binary label indicating the presence of depression.

5.2.1. Convolutional Neural Networks (CNNs)

A key component of DL architectures for grid-like data processing is CNNs. Convolutional Neural Networks [62] are among the most significant architectures in DL that are primarily used to analyse grid-like structured data. Their ability to understand both spectral and spatial patterns makes CNNs the most commonly applied DL-structured models for detecting depression. Examples of how CNNs are being used to detect depression include:

- **Speech analysis:** where prosodic features and spectral features suggestive of depressed speech can be extracted from audio spectrograms or time-frequency representations.
- **Facial emotion recognition:** where CNNs can be used to detect small variations in the way an individual looks at someone eye contact and facial expressions that indicate emotion.

Moreover, CNNs have also been used for EEG signal processing [63], [60] by taking EEG data in the form of spectrograms or topographic head maps and determining if abnormal brain activity is related to depression.

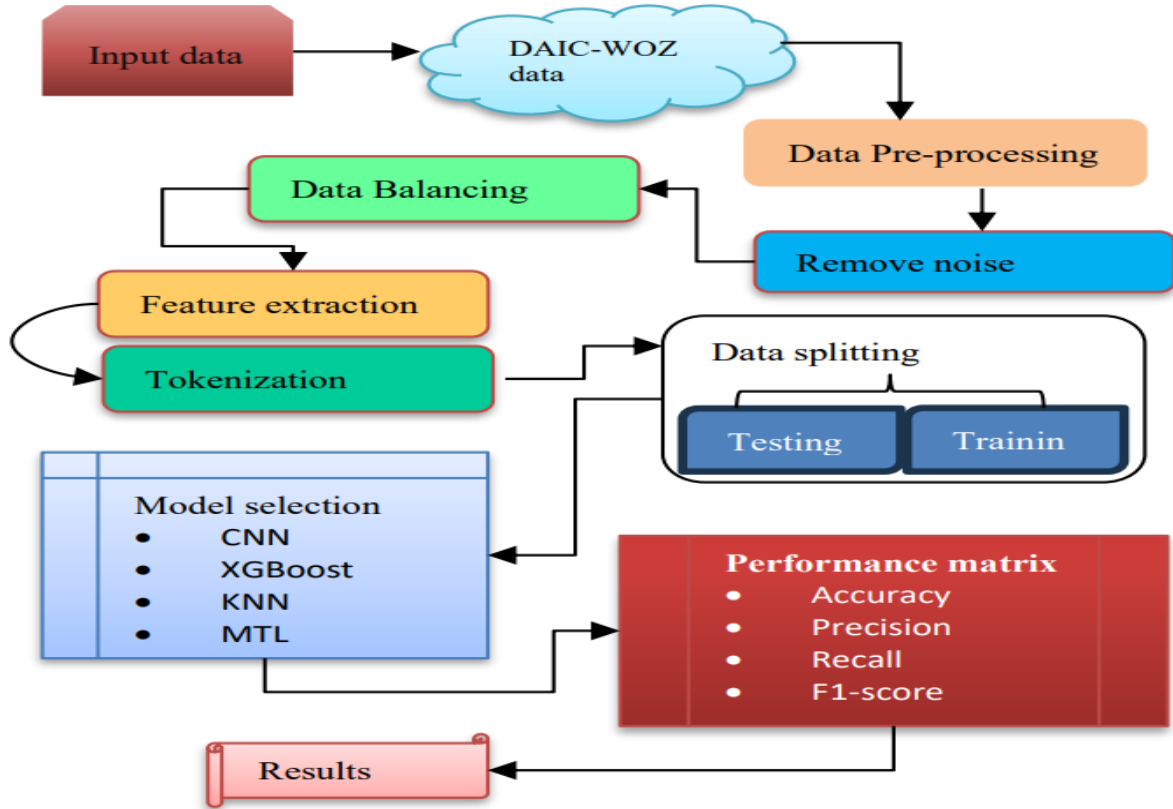


Figure 7: Process flow for automated depression detection in mental health systems [60].

The two core components of CNNs, convolution and pooling, allow CNNs to progressively learn how to detect different features from simple, more localized feature maps to much larger and more complicated features. These hierarchically structured levels of learning make CNNs perfectly proportioned for solving the above problems. Figure 8 illustrates a multimodal DL architecture combining a 1D CNN and a BiLSTM with an attention mechanism to integrate audio and textual embeddings for depression detection.

5.2.2. Recurrent Neural Networks (RNNs)

Recurrent Neural Networks [64] offer a unique approach to sequential input processing, enabling the identification of temporal dependency and both dynamic and evolving patterns over time. As a result, RNNs have gained significant traction in the area of depression analysis; they are particularly well-suited to addressing the fact that the symptoms of this condition frequently change throughout the course of a depressive episode. There are several important ways researchers use RNNs in this area, including:

- **Prosody analysis:** When analysing temporal aspects of an individual's speech throughout a depressive episode, researchers gain insight into how speech characteristics, such as prosody, pitch, and pauses, change as an individual transitions through the different stages of depression, as well as gain insight into the prediction of the likelihood of the existence of depression.

- **Linguistic analysis:** A linguistically-based method of assessment is based on analysing the frequencies of language patterns, such as syntax, lexicon, and sentiment, from social media posts, clinical notes, and interviews for variations associated with depressive symptomatology.

- **Physiological and wearable data:** Researchers can obtain physiological changes, such as changes in heart rate variability and changes in sleep quality, and wearable data such as time-stamped physiological signals, including heart rate variability and sleep quality from wearable devices, which provide researchers with an opportunity to identify the timing and duration of depression-related physiological changes.

RNNs have a very special ability. It can remember things from one moment to another and remember things that happened earlier. This allows them to analyze new information coming in at different points and predict what will happen next. Because RNNs can process information as it comes in and continue into the next moment, the sequence of events a person experiences develops a foundation of understanding for how depression develops over time. Figure 9 illustrates a typical RNN-based architecture for depression detection.

5.2.3. Long Short-Term Memory (LSTM) Networks

Long Short-Term Memory [66] networks are highly developed forms of RNNs, built to conquer the problem of vanishing

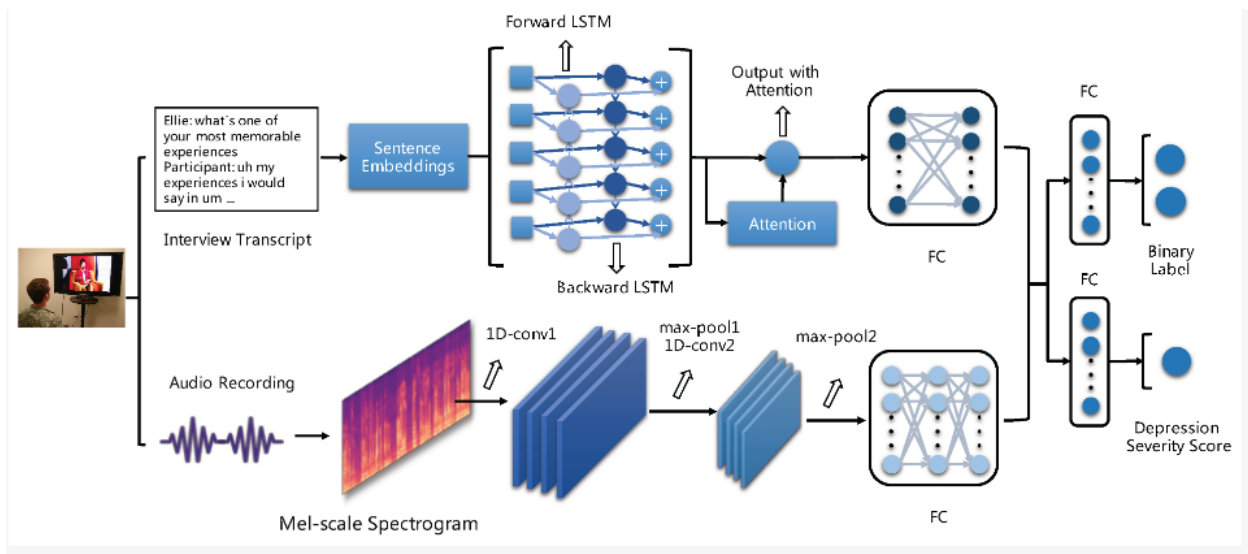


Figure 8: Multimodal architecture combining a 1D CNN and a BiLSTM with an attention mechanism for depression analysis [63].

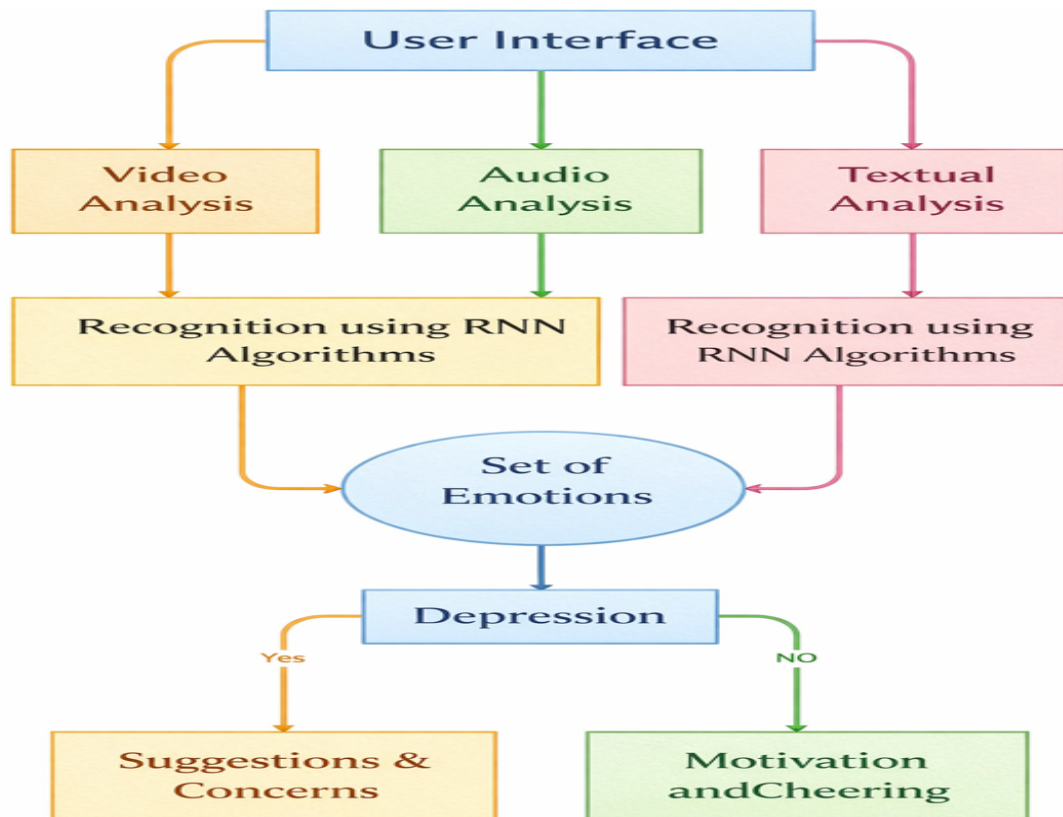


Figure 9: A RNN-based model for depression detection [65].

gradient and increase the length of time the RNN can remember temporal relationships within the data it receives. The model's inherent architecture grants LSTMs superiority over traditional RNNs at identifying multi-layered temporal mappings of the symptoms of depression over time.

The LSTM framework consists of several unique structural features to selectively remove past information and replace it with new and current information that includes: how much to remember (Input Gate), how much to throw away (Forget Gate), and what information to save at the point of producing an output (Output Gate). In this manner, LSTMs can more readily detect patterns that occur over time within a sequence of signals that might represent feelings of depression.

A common use case of this technology is processing long periods of collected speech, text, or signals from a wearable sensor. At the heart of all three examples is the challenge of making accurate predictions or creating monitoring tools that allow one to continue monitoring over an extended period of time. Figure 10 depicts a hybrid RNN-LSTM framework for depression detection [67]. Internal cell state and gating mechanisms (forget, input, and output gates) are the fundamental components of LSTMs, enabling them to selectively recall or forget data over time steps. With this feature, they can better capture the intricate temporal dynamics that are present in depression, which is a major improvement over ordinary RNNs for clinical decision support prediction.

5.2.4. Transformer-Based Deep Learning Algorithms

This review paper presents a detailed overview of how the new generation of transformer-based DL models [68] developed before the creation of LLM has played a major role in analyzing depression. They allow for the evaluation of the importance of all the words in a sequence at once, as well as the ability to understand the relationships between different words in the same sentence, regardless of how far apart they are.

Transformers have risen in prominence due to their capability to process large amounts of both text and multimodal information. In the field of mental health, the use of transformers for text-based applications has been employed to recognize patterns in text, such as linguistic characteristics and changes in emotions over time, such as sentiment, from multiple sources, such as clinical documents and Twitter, to identify whether an individual is likely to be experiencing any signs of depression. The multimodal use of transformers has also been developed to facilitate the combination of multiple data formats, such as physiological signals and speech transcriptions, into one predictive model for increased accuracy. Although there has been significant superiority shown by transformer-based models to date, three major barriers still exist: the requirement of a high level of computer processing power, large data requirements for sufficiently accurate training, and low interpretability of results from the complex structure of Transformer-based models. From the findings of multiple studies, it is evident that transformer-based models are at the forefront of a new paradigm for mental health informatics, as they will provide more precise detection of depression and more accurate stratification of risks from text-based and behavioural data.

5.3. Large Language Models (LLMs)

Large Language Models are a significant step in depression analysis because they leverage contextual embeddings and few-shot learning to better interpret meaning than standard methods. Their primary benefit is their capacity to analyze unstructured data, particularly conversational and clinical text, which makes them ideal for depression-related tasks including detection, categorization, and interpretation.

5.3.1. BERT and Domain-Specific Variants

Bidirectional Encoder Representations from Transformers (BERT) and its domain-specific equivalents have been and are widely used for depression analysis with encouraging results. The original BERT [69] model utilises a bidirectional transformer architecture to capture deep contextual and semantic information in text data, and that is effective at recognizing more subtle symptoms of depression in different text sources, including social media text or clinical interview transcripts. Previously discussed domain-specific variants of BERT include ClinicalBERT and MentalBERT, which pre-train on some corpus related to the interactions of clinic notes and mental health language. They utilize this pre-training to recognize linguistic features and psychological states related to depression.

ClinicalBERT enhances the identification of depression by optimising contexts relevant to medicine, whereas MentalBERT utilises discussions focused on mental health from platforms such as Reddit to collect colloquial and descriptive language associated with mental health discussions. According to the findings, the two versions of MentalBERT and the RoBERTa-based version outperform baseline models with better representations of emotion and embeddings of the language of mental health, emotional shifts, and symptom descriptions.

In conclusion, domain-adaptive pretraining and fine-tuning of BERT variants are crucial in the successful understanding and representation of the complexities of how depression is displayed and articulated in language while increasing sensitivity to emotion and context relative to traditional language models. Ultimately, accurate detection of depression from unstructured text will allow for earlier and more accurate detection of depression in the interest of timely and improved health. To highlight its function in accurate depression classification, Figure 11 shows the methodological pipeline of the BERT-SVM ensemble model.

5.3.2. LLaMA and Other Open-Source LLMs

Recently, Large Language Models [68] that are open source, for example, Falcon and Large Language Model Meta AI (LLaMA), have become increasingly relevant in the assessment of depression since they have the ability to understand and create language naturally. Meta AI's LLaMA focuses on resource efficiency and the ability to understand language accurately; therefore, LLaMA is considered an excellent choice for researching mental health issues where agencies are often limited by the amount of computational resources that they have access to. Fine-tuned versions of LLaMA [71] such as LLaMA2-7B

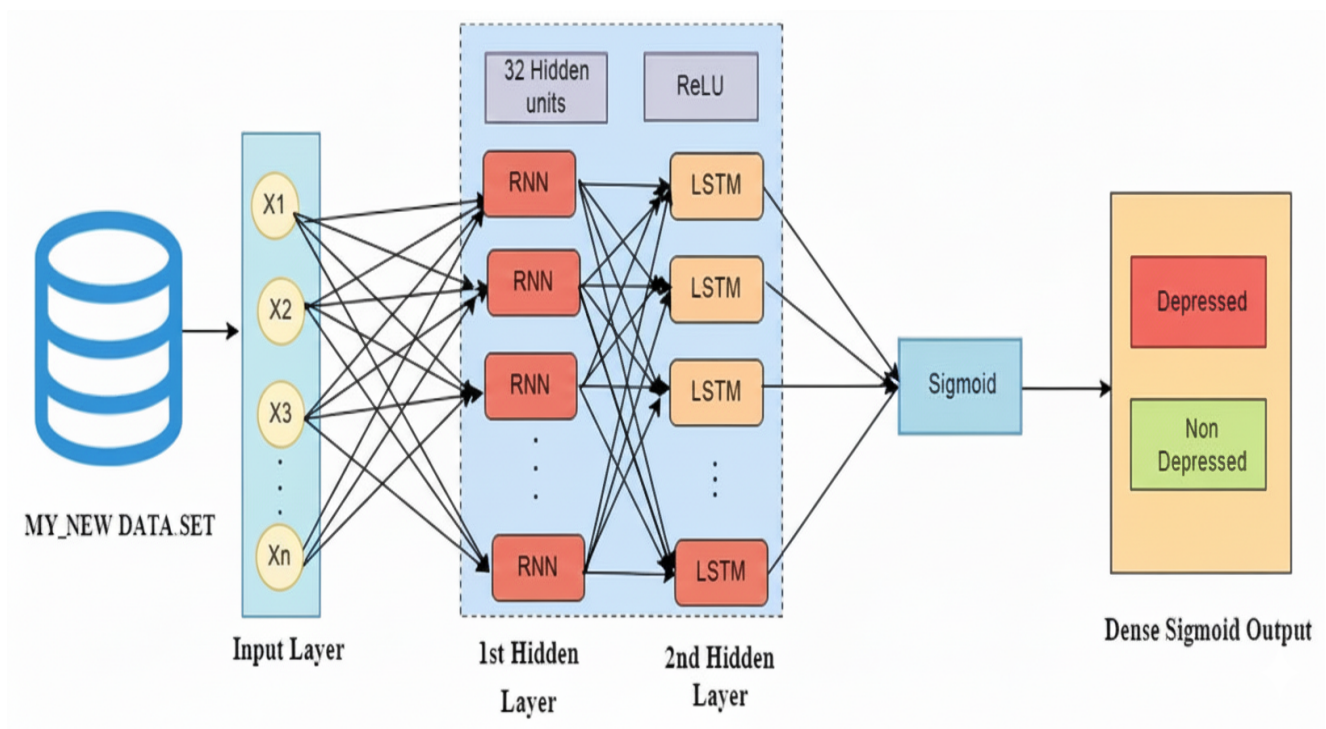


Figure 10: A RNN and LSTM-based hybrid Model [67].

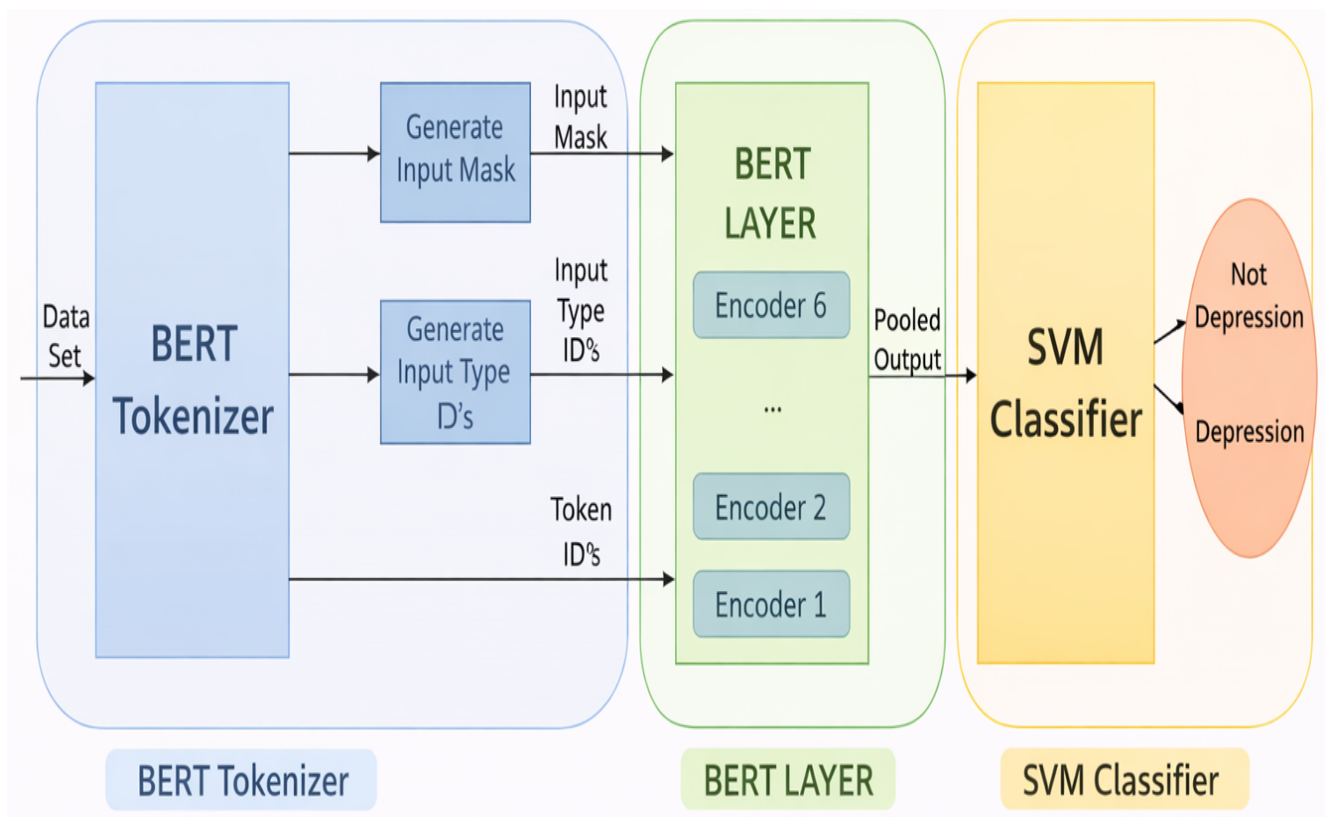


Figure 11: Methodological workflow of the hybrid BERT-SVM ensemble model for depression classification [70].

have demonstrated success in detecting depression through social media text with as much as 84% accuracy. This is a testament to LLaMA's ability to use more subtle, less obvious features in the language of individuals who suffer from mental health disorders. Figure 12 is a presentation of the LLaMA model, which incorporates psychological insights to improve diagnosis accuracy and is used to diagnose depression through speech analysis.

Falcon and other open-source LLMs have been studied similarly to detect depression using either clinical interview transcripts or conversation data. Research has also shown that LLMs produce distinct patterns of emotional expression and sentiment. A particular characteristic difference between LLaMA and other LLMs is that LLaMA generates more optimistic emotional responses; therefore, the language generated could affect how the user behaves, what the user is aware of, and how successful the user's therapeutic outcome will be. The transparent nature of open-source frameworks allows improved sensitivity of fine-tuned, open-source models to language cues related to depression.

5.3.3. *Learning Paradigms, Stability, and Practical Limitations of LLMs*

There are two primary paradigms for using LLM models in the analysis of depression: fine-tuning to create a task-specific corpus and using prompts to make inferences with zero or very little training data (prompt-based). While fine-tuning typically produces more consistent results across multiple datasets, prompt-based approaches provide users with an alternative means of conducting analyses with greater potential flexibility. However, due to their dependence on the wording of instructions and the manner in which they are framed, prompt-based methods are especially sensitive, which results in considerable variability among outputs produced by each model based on these aforementioned factors. As a result of this variable output from prompt-based methods, there are notable challenges to achieving reproducibility, as well as reliability within a clinical context, since even minor shifts to either prompts or texts may alter the model's behaviour significantly.

Although LLMs are very good at understanding and using language, they also have some shortcomings that are very important to be aware of in the field of mental health. There are multiple potential threats to using LLMs, such as inconsistencies in the output generated by the model, creating sounds-like-it's-reliable but in fact unverified information, and just the sheer volume and lack of transparency associated with the number of parameters that the model is trained on. Each of these concerns diminishes trust in the clinical output from LLM capabilities. The future of LLMs for a decision-support system is enormous, but there is still relatively little research regarding the use of LLMs in the mental health field, therefore limiting how easily we can incorporate LLMs into the functional component of clinical practice because of ethical considerations, privacy surrounding electronic health records, and the lack of established guidelines to evaluate and implement LLM technology.

Recently, benchmark-type evaluations were done comparing many different LLM architectures, such as BERT-style, RoBERTa

variants, and generative LLMs, on depression-related datasets using the same evaluation criteria. The findings of these studies indicate that fine-tuned language models yield more consistent and reproducible results than prompt-based language models, but the results vary greatly in performance across datasets, tasks, and other evaluation processes.

The contemporary literature suggests that LLMs are more supportive tools for the analysis of information and the screening of symptoms, as well as assisting clinicians, as opposed to diagnostic automation systems. Therefore, caution is required in the validation of LLMs prior to widespread clinical use.

5.4. *Hybrid and Ensemble Approaches*

For the purpose of improving forecast accuracy and resilience, hybrid approaches that integrate ML, DL, and LLM models are becoming increasingly popular. By utilizing the complementary characteristics of various paradigms, which include the structured-data efficiency of ML, the intricate pattern identification of DL, and the unstructured-text understanding of LLM models, these ensembles present predictions that are more adaptable and dependable than any single method alone.

5.4.1. *Feature Fusion Models*

Feature fusion models combine information from different data sources to increase depression prediction performance. These models typically deal with two kinds of inputs: unstructured data, like text, images, or speech, which need automated feature extraction using DL methods like CNN or RNN, and structured data, which comes from conventional assessments, surveys, or clinical measurements [73]. In traditional fusion approaches [74], ML algorithms such as SVM, RF, and GBM are used to identify meaningful patterns in structured features, whereas DL models extract high-level representations from unstructured data. Compared to single-source models, feature fusion models can collect complementary information by merging these diverse feature sets, allowing for more precise and reliable predictions. Figure 14 exhibits multimodal speech-text feature fusion strategies for depression detection.

When applied together, classical ML and DL provide better depression assessment predictions and a wider range of possible uses than when used separately. Hybrid and fusion-based approaches enhance classification accuracy, tackle issues like inter-subject variability and data imbalance, and offer a more thorough knowledge of depression.

5.4.2. *Ensemble Voting Systems (Stacking SVM with LLM Embeddings)*

In depression analysis, ensemble voting systems [76] have proven to be a potent technique, particularly for enhancing prediction robustness and accuracy. One of the most effective ensemble strategies [77] involves combining SVM with LLM embeddings, resulting in a hybrid approach that capitalises on both conventional ML and advanced natural language processing (NLP) techniques. In this approach, the SVM acts as the primary classifier to classify decision boundaries from features extracted from structured data. In parallel, the LLM embeddings, such

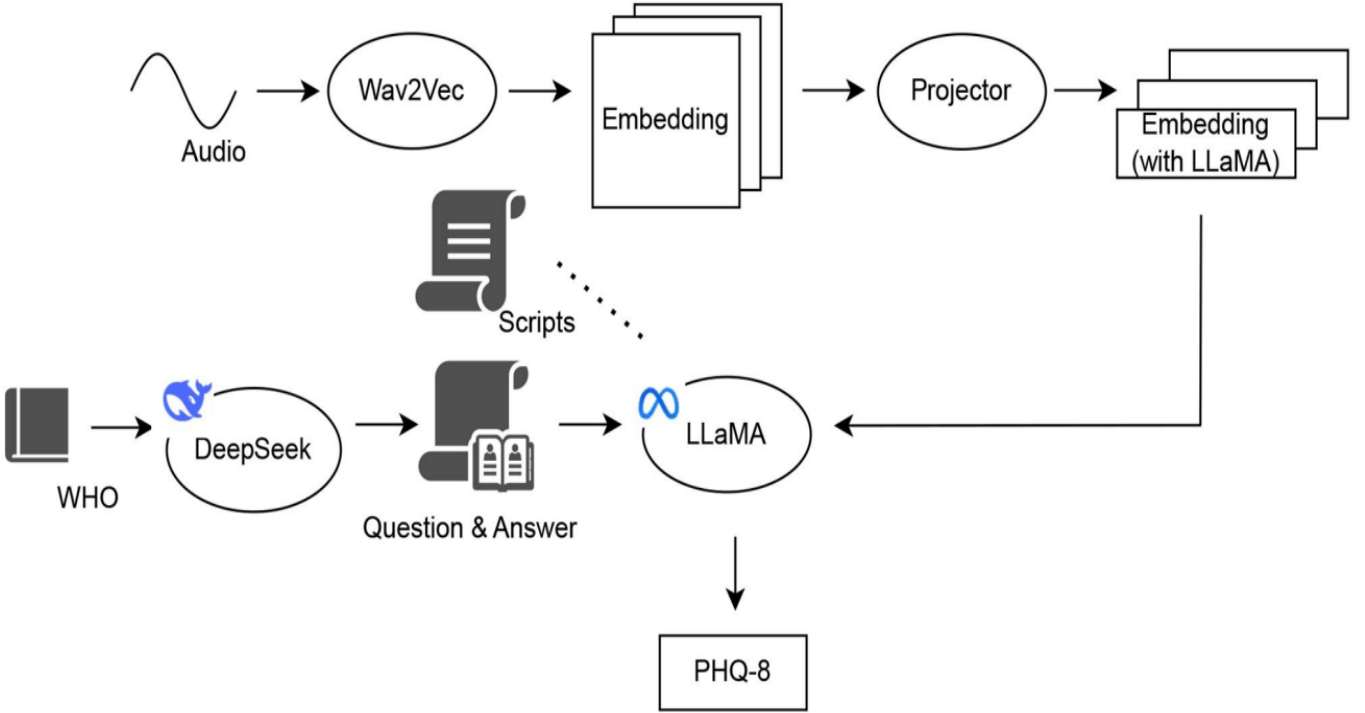


Figure 12: LLaMA for depression detection in speech using psychological insights [72].

as pre-trained GPT and BERT, capture richer, contextualised factors from unstructured data such as text or voice. The ensemble framework makes the prediction from each of the models, which will make a vote to deliver a final prediction. This strategy enables the individual models' strengths to complement each other, thereby improving overall performance.

By stacking SVM with LLM embeddings for depression analysis, the system effectively combines the interpretability of ML algorithms with the advanced, context-sensitive capabilities of DL models. Through the integration of multiple predictive models, this ensemble approach has demonstrated superior performance in addressing challenges such as data heterogeneity, sparsity, and the complexity of depression, providing clinicians with a more accurate tool for diagnosis and detection. In another study [75], a stacked ensemble model with feature importance produced from RF is used to present the method of depression detection that is presented in Figure 13.

5.4.3. Multimodal Fusion (Text, Audio, Video)

A common method for analysing and predicting depression is through Multimodal Fusion [78], which combines multiple types of data, including text, audio, and video inputs, to create a more accurate prediction. With this technique, information from text-based sources, such as Clinical Notes, Social Media, and recorded conversations, will become the text-based features. The audio modality comprises features such as prosody, tone, and speech patterns, which are assessed using signal processing models followed by DL applications. Visual data modelling focuses on extracting facial expressions and body language using computer vision techniques, often employing some

form of CNN to identify subtle emotional information.

Fusion of modalities is usually carried out via early fusion, which combines features before the classification stage, or late fusion, which combines model predictions after classification. This multimodal model takes advantage of each modality while mitigating complexities in individual differences, inconsistent speech or expression, and contextual ambiguities. Overall, this analysis improves the ability of the system to detect depression and yields a more accurate, reliable, and clinically relevant diagnostic and monitoring tool. Voice-text feature fusion depression detection strategies are shown in Figure 14.

6. Ethical, Regulatory, and Clinical Deployment Considerations

Machine Learning, Deep Learning, and Large Language Model-based systems have great potential for detecting depression. However, its application in clinical practice creates serious ethical, legal, and clinical difficulties. Because mental health data is so sensitive, algorithms may mispredict or misinterpret it, delaying therapy or resulting in improper patient management. As a result, the implementation of these technologies necessitates strong ethical principles, compliance with legal requirements, and explicit human monitoring. To appropriately transfer these models from research settings into clinical practice, careful incorporation of these protections is necessary.

6.1. Ethical Risks and Privacy Concerns

An essential ethical consideration in automated depression analysis is the protection of human privacy and the responsible

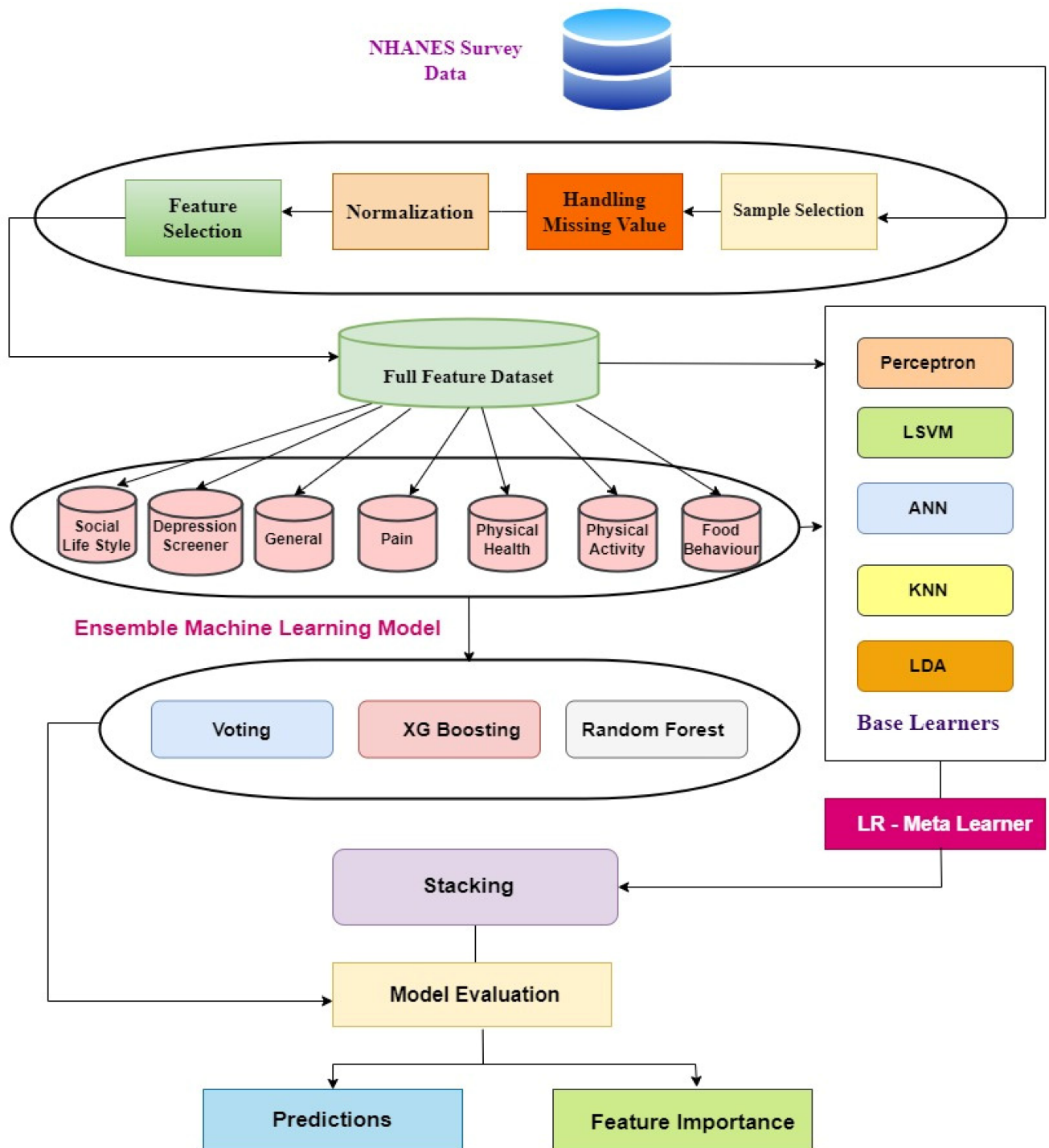


Figure 13: Depression detection using stacked ensemble and RF feature importance [75].

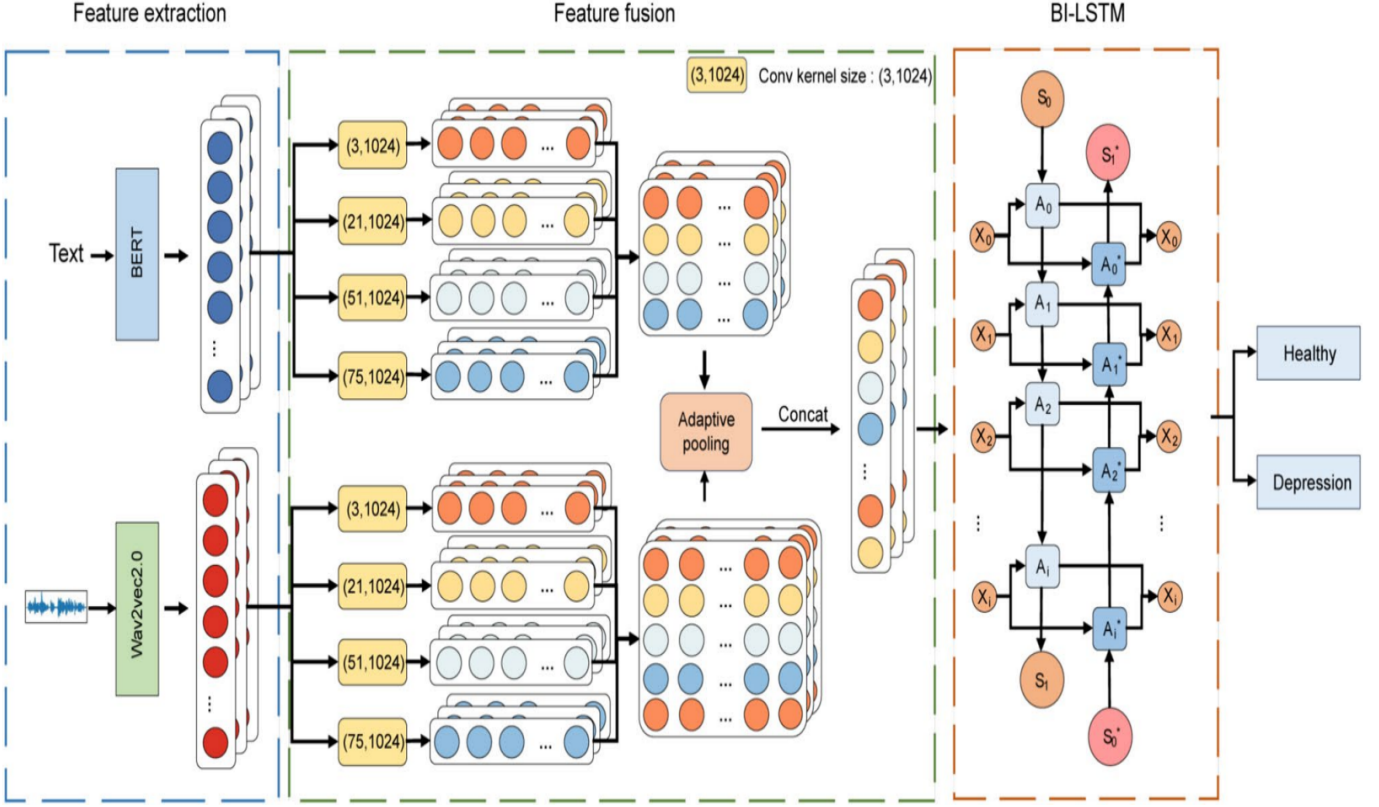


Figure 14: Overview of depression detection methods based on multimodal fusion of voice and textual features [73].

utilization of sensitive mental health information. Data, typically gathered from surveys, electronic medical records, and social media platforms, often includes personally identifiable and emotionally sensitive information. Consequently, stringent ethical norms are imperative to avert faulty processing, insufficient anonymization, or illegal use, which may result in significant privacy violations and societal detriment [79].

In addition, ethical issues with respect to bias and fairness further complicate the ethical deployment of these systems. For example, models developed from demographically imbalanced or culturally narrow data sets may not provide accurate predictions for minority or underrepresented populations, thereby perpetuating the existing inequities related to the assessment of depression and access to care [80]. To limit this training to such document types will diminish the LLM’s ability to develop generalised capabilities to support those in a wide variety of languages and cultures. On the other hand, LLM-generated text may look correct and sound authoritative; however, some of the content is incorrect and unverifiable, which is referred to as hallucination [81]. Hallucinations present great challenges regarding the appropriateness of LLM-generated text used for analysing and interpreting language related to mental health, and there needs to be a proper level of validation and ethical approval given to LLM-generated text utilised in mental health assessments before the release of any LLM in the public domain.

6.2. Regulatory and Compliance Challenges

From a regulatory perspective, AI-based depression testing applications or AI systems that provide clinical decision support continue to be classified as health-related or medical technologies and, as such, are subject to considerable regulation. Most regulatory agencies have typical frameworks that address data protection, transparency, audit trail, and risk assessment across the life cycle of an LLM’s development and deployment [82]. However, most ML or LLM-based depression detection applications remain at the stage of research and experimental development and do not have specific regulatory approvals to be adopted in clinical practice.

The complexity and opacity of advanced models present additional regulatory challenges. Traditional ML models, like LR and DT, provide fairly interpretable methods for decision-making, so they’re easier to understand. In contrast, DL architectures and LLMs have a very black-box approach, making it difficult for practitioners to understand how predictions have been developed. Also complicating the reliability and reproducibility of these systems are sensitivity to the training data sets, prompt formulation, and model updates, all of which pose significant barriers to large-scale clinical deployment. These factors also impede alignment of these systems with existing regulatory standards, which are necessary steps to achieve clinical validation.

6.3. Risks of Fully Autonomous Depression Screening

There are substantial clinical concerns associated with using completely automated systems as diagnostic or screening tools on their own. People who are mistakenly thought to be depressed may incur needless suffering and shame as a result of false-positive predictions, which can force them to be referred to secondary treatment without a valid reason. On the contrary, wrong diagnoses can delay or even prevent people who are suffering from real depression from getting the help they need. These dangers are heightened when using social media data for analysis because of issues like dataset noise, language heterogeneity, and domain shifts that might occur across various institutions or geographic populations.

LLM-based systems raise extra difficulties due to output instability across model versions and susceptibility to slight changes in input phrasing. However, these models fall short when it comes to actual clinical reasoning and awareness of context, even when they show impressive language abilities. Thus, there is a lot of warning in the current research about using completely autonomous AI systems to diagnose mental problems in patients [83].

6.4. Decision-Support and Human-Centred Deployment Frameworks

To address ethical and clinical risks, contemporary research increasingly advocates for human-centred deployment strategies in which ML and LLM systems function as decision-support tools rather than autonomous decision-makers. Within these frameworks, commonly described as human-in-the-loop or clinician-in-the-loop approaches, AI models assist by identifying risk patterns, summarising patient information, or prioritising cases for clinical review, while final decisions remain under professional medical judgment [84].

Hybrid AI deployment approaches provide ongoing monitoring, validation, and rectification of algorithmic results, thus improving reliability and accountability. These methods are especially beneficial in impoverished areas and culturally diverse settings, where physicians' contextual understanding and skills are crucial for precise evaluation. Using decision-support frameworks is a useful and moral way to use ML and LLM technologies in mental health care and depression analysis because they combine computer efficiency with human oversight.

7. Result Analysis

7.1. Evaluation Metrics

Table 4 illustrates the primary assessment criteria used to evaluate the models for depression analysis. It defines and explains how to calculate AUC, F1-score, recall, precision, and accuracy [85]. These metrics provide a solid and useful foundation for evaluating models, particularly in datasets that are imbalanced and where using just one measure could be deceiving. In particular, they make it easier to get an extensive overview of how accurate, sensitive, reliable, and overall effective the model is in finding favorable instances. Also, Table 5

reviews a range of ML and LLM investigations (2023–2025) focused on depression detection, classification, and prediction using varied datasets and approaches. The cited works encompass social media analysis (e.g., Twitter, Reddit, Facebook), clinical and demographic data, and multimodal sources integrating text, audio, and facial expressions.

Zulfiker et al. [23] reported a Bangladesh-based study that predicts depression using socio-demographic and psychosocial survey data from 604 individuals. Using 30 variables collected from the Bangladesh depression checklist, the researchers utilized AdaBoost with SelectKBest feature selection to improve prediction power. The model obtained very good performance, with a high accuracy of 92.56%, AUC of 0.96, and other strong metrics (precision and F1 score). With this data demonstrating that organised self-reported mental health data can provide valid measures of mental health, it is essential to note that, because it solely relied upon self-reports, there are limitations associated with the external validity of the findings due to social desirability bias, under-reporting, and subjectivity involved in interpreting each question. Additionally, while the sample size is adequate for exploration, the sample remains somewhat small with respect to clinical use. Nonetheless, despite these limitations, the study provides a strong argument that improved classification of depression in electronic survey data may be achieved using ML.

The study by Islam et al. [91] investigated depression identification through Facebook comments, evaluating more than 7000 postings categorized as depressive or non-depressive. A DT model was utilized, enabling interpretability but limited sophistication compared to recent NLP techniques. This study's findings provided some encouraging results, as indicated by an F1-score of 74%, but had very variable results about both precision and recall. These inconsistencies are likely due to the data itself having many unusual patterns of language usage. In addition, the noisy nature of many social media messages, such as sarcasm, the casualness of the language, and the ambiguity of the emotional intent, created obstacles for the system to accurately identify depressive symptoms. Additionally, the model failed with the domain transition between actual depressive writing and informal internet conversation. Accuracy ranged significantly between 60–80%, showing instability problems. The study demonstrates the potential of social media language for mental health identification, while also illustrating the limitations of employing basic models on messy, user-generated information.

Cao et al. [92] examined 47 studies that used ML algorithms to identify depression on Facebook, Reddit, Twitter, and other social media sites. Their findings showed that there were significant variations in assessment techniques, datasets, and methodologies between studies, which made it difficult to compare studies and avoid doing a formal meta-analysis. The evaluation concluded that the majority of studies only used Twitter and other datasets in English, which limits the applicability of the results to other platforms and people who speak languages other than English. It also demonstrated that demographic biases, errors in methodology, and different data sources all had a negative impact on model performance. There has been a lot

Table 4: Common Evaluation Metrics Used in Depression Analysis

Metrics	Description	Formula	Studies
Accuracy	ML Accuracy is defined as the number of correctly predicted occurrences divided by the total predicted occurrences. The prediction cannot include just the True Positives (TP) and True Negatives (TN), but should include False Positives (FP) and False Negatives (FN). Therefore, while the accuracy metric may give an accurate estimate of the ML model's ability to predict imbalanced data, precision and recall should also be used to verify the results from the accuracy metric.	$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100$	[86]
Precision	Precision is the metric that is part of the ML community that indicates the percentage of positive predictions that the model made that were actually positive, to the total number of positive predictions that the model made. The precision metric indicates how well the model has been able to accurately identify the positive examples and the number of positive predictions that have produced false positives.	$\text{Precision} = \frac{TP}{TP + FP} \times 100$	[87]
Recall	Recall, also referred to as Sensitivity or True Positive Rate, is calculated by taking the ratio of True Positives that were found, divided by the total positives in the dataset. This metric helps to measure the ability of the algorithm to find the correct relevant instances without missing any. False Negatives are critical to identify since they have the potential for harm.	$\text{Recall} = \frac{TP}{TP + FN} \times 100$	[88]
F1-Score	The F1 score summarises the performance of an ML algorithm by using both Precision and Recall metrics and combining them to produce a single figure. By combining these two metrics together, the F1 score creates a way to measure how well the ML algorithm was able to find all of the True Positive instances and account for them while also identifying any False Positives and False Negatives. To find the F1 score, you will take the Harmonic Mean of Precision and Recall.	$F1 = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \times 100$	[89]
AUC Score	The Area under the Receiver Operating Characteristic Curve (AUC-ROC) measures a model's discrimination performance. The ROC curve features the True Positive Rate(TPR) plotted against the False Positive Rate(FPR). AUC is a higher level of classification performance.	$\text{FPR} = \frac{FP}{FP + TN}$	[90]

Table 5: Analysis of experimental results from selected research papers (2023–2025) on ML and LLM approaches for depression analysis.

Ref.	Application	Dataset	Preprocessing and data augmentation methods	Model	Results	Limitations
Zulfiker et al. [23]	Depression prediction using socio-demographic and psychosocial features	Survey of 604 participants (Bangladesh), 30 variables from BDC	✓	AdaBoost with SelectKBest	Accuracy: 92.56%, AUC: 0.96, Precision: 95.77%, F1: 93.79%	Self-report scale; no biological data; limited sample size
Islam et al. [91]	Depression detection from Facebook comments	7,145 Facebook comments (4,149 depressive; 2,996 non-depressive)	✓	DT	F-measure: 0.74; Precision: 0.57–0.63; Recall: 0.40–1.00	Doesn't identify users; accuracy 60–80%; limited LIWC attributes
Cao et al. [92]	Systematic review of ML for depression detection on social media	47 primary studies from Twitter, Reddit, Facebook, etc.	✓	Synthesis of SVM, RF, BERT, etc.	Platform (Twitter-centric); language (English-centric) issues	English-only studies; heterogeneity prevents meta-analysis
Vu et al. [93]	Depression prediction using health and demographic data	NHANES 2013–2014 (5,372 participants), PHQ-9 ≥ 10	✓	XGBoost	Accuracy: 0.69; Sensitivity: 0.68; Specificity: 0.71; Precision: 0.70; AUC: 0.69; F1-score: 0.69	Cross-sectional design; class imbalance; recall bias
Zhao et al. [94]	Depression risk prediction in COPD patients	CHARLS 2018 (2,921 COPD patients), CESD-10 scale	✓	XGBoost	Test: Accuracy 67.50%; AUC: 0.747; Val: Accuracy: 70.63%; Sensitivity: 59.05%; F1 score: 63.17%	Retrospective; missing COPD variables; limited population
Bokolo et al. [95]	Depression detection from tweets using ML and Transformers	Sentiment140 (632k labeled tweets for depression)	✓	RoBERTa	Accuracy: 98.1%; Precision: 0.99; Recall: 0.99; F1 score: 0.98; Mean Accuracy: 97% (10-fold CV)	Repurposed dataset; English only; no multimodal validation
Rashid et al. [96]	Comprehensive evaluation of LLMs for depression detection and classification	Compilation of studies from clinical notes and social media	✓	LLM variants: GPT family and fine-tuned BERT	LLMs promising for various tasks; embedding models useful for downstream ML	Publication bias; rapid model changes; dataset heterogeneity; ethical and privacy concerns
Sadeghi et al. [97]	Multimodal depression detection using LLMs and facial expressions	Multimodal datasets: text diaries, clinical interviews, and video features	✓	Multimodal fusion (LLM for text and vision model for facial expressions)	Multimodal fusion outperformed text-only or face-only models	Complexity; ethical and privacy concerns; generalizability across recording settings

Ref.	Application	Dataset	Preprocessing and data augmentation methods	Model	Results	Limitations
Lawrence et al. [98]	Review of LLM applications for mental health assessment, education, and screening	Survey of articles and application instances (review)	✗	Evaluates various LLM approaches	Summarizes promising task performance while highlighting risks (safety, hallucination, regulatory gaps)	Fast-evolving field; review may become outdated quickly; limited clinical deployment evidence
Ji et al. [99]	Evaluate LLMs to detect depression or anxiety symptoms in patient portal conversations	Secure patient messaging or clinical communication corpus (multi-center)	✓	Fine-tuned LLMs (GPT-series or similar) with optimum prompt approaches	LLMs work effectively when prompt-engineered; few-shot and persona strategies boosted sensitivity	Privacy constraints; potential for hallucination; required for full safety testing before clinical use
Wang et al. [100]	Investigation of LLM performance and biases in depression detection across demographic groups	Language corpora stratified by demographic groups for fairness analysis	✓	LLMs (various transformer models fine-tuned)	LLMs show potential but performance varies by language or demographic; risk of reduced accuracy for underrepresented groups	Fairness and generalization issues; need for demographic-aware validation
Baydili et al. [101]	DL framework combines pre-trained language models and feature selection for depression detection	Clinical notes and social media text corpora (research employs mixed sources)	✓	Fine-tuned pre-trained LM (e.g., BERT family) with specified downstream layers	Competitive classification performance; shows coupled feature selection enhances robustness	Reproducibility across platforms; privacy concerns when using clinical notes
Kurniadi et al. [102]	Detect depression from social media text using transformer models	Collected social media posts (Twitter and Reddit samples)	✓	RoBERTa fine-tuned; outperformed BERT	Notable accuracy and F1 improvements vs old models; highlights usefulness of pre-trained transformers	Language and domain shift sensitivity; tiny labeled datasets restrict fine-tuning stability
Nusrat et al. [103]	Use tweets to figure out what kind of depression someone has (bipolar, serious, postpartum, etc.)	Twitter corpus labelled for different types of depression (using lexicon and annotation techniques)	✓	Transformer-based classifier fine-tuned with BERT	Better at telling the difference between several classes than previous methods; F1 changes with class	Lexicon labelling adds noise to the labels; hard to tell clinically various illnesses apart only from text

Ref.	Application	Dataset	Preprocessing and data augmentation methods	Model	Results	Limitations
Vu et al. [104]	Use ML to predict depression disorder using multimodal clinical	Multi-centre clinical datasets (questionnaires and EHR characteristics) with reasonable sample size	✓	Ensemble tree model (e.g., XGBoost) with interpretability techniques	Show that AUROC and balanced accuracy indicate clinically relevant discrimination in validation sets	Need for external validation; varying data quality between centres; risk of confounding
Devaguptam et al. [105]	Early detection of depression using sequential Reddit posts	Reddit postings (chronological sequences from users) organized for early warning detection	✓	DeBERTa fine-tuned classifier (best of transformer variations tested)	Significant improvements over classical baselines; greater early detection when sequence context is kept	Reddit user demographic bias; label noise from distant supervision; temporal generalisation problems
Jamali et al. [106]	Using ML to predict depression from conventional health and questionnaire datasets	Publicly available questionnaire datasets	✓	XGBoost with SHAP interpretability	Good discrimination for risk prediction (AUROC >0.7); SHAP identified key predictors	Cohort-specific factors; low generalisability across groups; reliance on self-report labels
Gadzama et al. [107]	Systematic review of ML and DL techniques for detecting depression on social media	Aggregated studies from Twitter, Reddit, and Facebook (different sizes)	✓	Various; BERT and transformer fine-tuning highlighted as top performers	Improved performance with transformer models; varied metrics across experiments (AUROC varies by dataset)	Heterogeneous datasets; English-centric approach; inconsistent evaluation and labelling; risk of bias in primary research
Cao et al. [108]	Multimodal depression consultation dataset for AI-based diagnosis and benchmarking LLMs	PDCH dataset, 100 real clinical consultations from Beijing Anding Hospital (audio, text, and HAMD-17)	✓	GPT-4o-mini-audio-preview	F1 score: 0.407 (text and audio); multimodal slightly improved performance; realism validated vs. MODMA	LLMs underperform clinicians; dataset limited to 100 samples; long recordings increase difficulty; Chinese-only; limited emotional diversity

Ref.	Application	Dataset	Preprocessing and data augmentation methods	Model	Results	Limitations
Teferra et al. [109]	Automated depression screening using zero-shot LLMs on clinical transcripts	DAIC-WOZ dataset (text, audio, video from 189 interviews; PHQ-8 scores)	✓	GPT-4o (zero-shot, RISEN prompt engineering)	Accuracy: 75.9%, F1: 0.74 (binary PHQ-8 scoring); best across emotional and cognitive items	Single dataset limits generalizability; text-only (no multimodal cues); model interpretability challenges; data imbalance; ethical concerns on fine-tuning
Lan et al. [110]	Depression detection from social media posts using LLM	Reddit & Twitter depression-related posts	✓	LLaMA-based LLM with DSM-guided depression recognition (DORIS model)	Outperformed standard LLM baselines on binary depression classification (improved F1-score and accuracy)	Limited to textual modality (no multimodal fusion); lacks real-time adaptability; potential dataset bias; limited generalizability
Li et al. [111]	Depression recognition from spoken language integrating psychological knowledge	DAIC-WOZ dataset (189 participants with audio and transcripts)	✓	LLaMA fine-tuned with Wav2Vec 2.0, LoRA, and knowledge injection	MAE: 5.35; RMSE: 6.42 (text and knowledge injection, best performance)	Limited GPU resources; used LoRA; small dataset size; no visual modality; difficulty generalizing audio-to-text knowledge transfer
Sadegue et al. [112]	DL framework combining pre-trained LMs and feature selection for depression detection	Clinical notes and social media text corpora (mixed sources)	✓	Fine-tuned BERT family with downstream layers	Competitive classification; coupled feature selection enhances robustness	Reproducibility across platforms; ethical and privacy concerns
Tadesse et al. [113]	Depression detection from social media using transformer models	Collected Twitter and Reddit posts	✓	RoBERTa fine-tuned; outperformed BERT	Notable accuracy; F1 improvements vs traditional models; highlights transformer usefulness	Language and domain shift sensitivity; small datasets limit fine-tuning stability

of progress, but there are still some big gaps, especially when it comes to dealing with cultural and linguistic diversity, setting up common evaluation methodologies, and making it easier to move data between platforms.

With the goal to forecast the symptomatology of depression using NHANES health data and PHQ-9 scores, Vu et al. [93] developed an ML model. The study used the XGBoost algorithm to find trends in demographic and health-related variables based on information from more than 5,000 people. Structured health data may aid in the identification of persons prone to depression, as the model demonstrated modest predictive accuracy with a sensitivity of 0.69 and a specificity of 0.71. But compared to other text-based models, its performance was poor. The study was constrained by its cross-sectional methodology, the possibility of recall bias in self-reported data, and demographic imbalance, which may influence the model’s robustness and generalizability.

Applying the 2018 CHARLS dataset, Zhao et al. [94] determined the likelihood of depression in COPD patients. The CESD-10 scale was utilized to detect depression, and 11 important predictors were found using an XGBoost model with LASSO feature selection. Results showed that the model could predict depressive symptoms in this subgroup to a decent degree; test accuracy was 67.5%, AUC was 0.747, and validation accuracy was 70%. Nevertheless, it faces challenges in catching all genuine cases of depression due to its lower sensitivity and F1 score. Due to its retrospective design, low generalizability, and absence of COPD-specific characteristics, the study has certain limitations. Overall, the results point to the potential of organized medical data to help forecast mental health risk, albeit more work has to be done.

Bokolo et al. [95] developed a RoBERTa-based model on more than 632,000 tagged tweets to find signs of sadness, and it got one of the best results in the dataset. The research achieved 98.1% accuracy, 0.99 precision, 0.99 recall, and a 0.98 F1 score using meticulous text preprocessing, TF-IDF feature extraction, and the power of transformer models. The results highlight the promising future of transformer-based models for evaluating online discourse for the purpose of depression categorization. However, the study’s therapeutic value is diminished by its reliance on reused datasets, use of English-only tweets, and absence of multimodal context. Another issue is that sampling bias is still a possibility, and tweets could not necessarily reflect people’s actual emotional states. Although there are several limitations, the study nevertheless shows that transformer models are very good at detecting depression.

Rashid et al. [96] examined LLM-based algorithms that use clinical notes and social media data to detect depression. According to the results, fine-tuned LLMs outperform more traditional ML models when it comes to picking up nuanced language cues. However, their practical application is limited due to publication bias, rapid technological change, inconsistent reporting, and ethical problems, particularly addressing clinical data privacy. The review emphasizes the necessity of transparent evaluation frameworks, safer deployment techniques, and setting benchmarks for trustworthy use in mental health analysis.

Sadeghi et al. [97] have created a multimodal system that uses facial expressions, text, and speech to identify depression. By utilizing a fusion architecture based on a broad language model, the system was able to successfully combine these inputs and extract verbal and non-verbal cues representing mental states from recordings of clinical interviews. This strategy outperforms text-only approaches when it comes to evaluating depression. Also, the researchers found that multimodal techniques boost performance, especially in recognizing subtle emotions. However, face and audio data privacy problems make these systems computationally expensive, ethically sensitive, and difficult to deploy. Cultures interpret facial expressions differently, limiting global generalizability. The study suggests multi-modal LLMs may enhance mental health diagnosis in the future, despite current obstacles.

Lawrence et al. [98] discussed the applications of LLMs in the fields of mental health screening, education, and patient assessment. The study doesn’t offer a new paradigm, but it does review the literature on the topic of how LLMs might aid in diagnosis, communication with patients, and the identification of mental health crises. The authors draw attention to important pros, such as scalability, speed, and reduced costs, but they also raise serious cons, like hallucinations, privacy problems, and a lack of evidence of safety from clinicians. The review presents a fair and impartial analysis, highlighting the importance of more uniform assessment frameworks and stricter regulatory standards.

Ji et al. [99] explored how LLM can be optimized to detect depression and anxiety in secure patient-portal communications. Using multi-center clinical message data, the study shows that prompt engineering, especially persona-based and few-shot prompting, can improve LLM sensitivity in mental health screening. The findings suggest that prompt-optimized or fine-tuned LLMs can be effective, although privacy concerns, hallucination risks, and the need for strong safety validation remain major barriers to clinical use.

Wang et al. [100] utilized demographically-split language corpora to study demographic bias in LLM-based depression identification. The study demonstrated that variations in representation and language patterns led to disparities in model performance between groups. Fairer comparisons were made possible by normalization and demographic stratification, although the accuracy of underrepresented populations suffered. These results highlight the importance of training that takes demographics into account, together with stricter fairness criteria and improved methods of rebalancing.

Baydili et al. [101] introduced a system for depression diagnosis using DL, which integrates language models that have already been trained with feature selection methods applied to clinical notes and social media data. To enable the creation of competitive classifiers, the preparation pipeline incorporates text cleaning, feature selection, and, on occasion, data enrichment. The results demonstrate that robustness and interpretability can be enhanced by integrating transformer-based representations with effective feature selection. Nevertheless, there are still significant constraints, such as worries regarding cross-platform repeatability and the ethical utilization of clinical data.

Applying models based on transformers, Kurniadi et al. [102] investigated the possibility of detecting depression on Reddit and Twitter. The study enhances performance by applying basic NLP preprocessing, data balancing, and modest augmentation. By demonstrating tangible improvements in accuracy and F1 score, RoBERTa demonstrated the superiority of lexicon-based labeling over previous transformer models. Limitations associated with self-reported data, language variety, domain shift, and instability during fine-tuning due to noisy and platform-specific labels are also highlighted in the study.

Nusrat et al. [103] used Twitter data pertaining to bipolar disorder, postpartum depression, and major depressive disorder to train a transformer-based classifier. The model outperformed previous methods in detecting small changes among depression categories after undergoing normal text preprocessing and data augmentation. Because clinical diagnoses and self-reported or inferred labels do not necessarily coincide, lexicon-based labeling has the potential to introduce noise and reduce the model's reliability and clinical relevance.

Vu et al. [104] developed an ML model to anticipate the intensity of depression based on structured electronic health record factors and responses to questionnaires. The processed data was utilized to train an Ensemble-XGBoost classifier with explainability tools after data cleansing, feature engineering, and selective augmentation. Based on the model's balanced accuracy and strong AUROC, it appears to have good generalizability. The authors point out that dependability may be impacted by variations in data quality and confounding variables across clinical contexts; external validation is still required.

8. Comparative Analysis of Machine Learning, Deep Learning, and Large Language Model Approaches

Mental health research has progressively utilized several computational methodologies for the diagnosis and analysis of depression. To evaluate ML, DL, and LLM methods, this review takes a task-oriented and clinically informed comparison approach rather than providing a purely descriptive survey. The analysis is organized around important aspects that directly influence clinical adoption in practice, such as task fit, data characteristics, clinical risk exposure, model transparency, and deployment feasibility.

This review aims to comprehensively evaluate the relative strengths, limits, and practical usefulness of ML, DL, and LLM techniques for depression assessment using a Clinical Readiness and Risk-Aware Comparison Framework (CRRCF).

8.1. Comparison Framework and Evaluation Dimensions

The proposed framework assesses ML, DL, and LLM approaches across four clinically relevant dimensions: (i) intended task category, (ii) data modality and sample efficiency, (iii) clinical risk and decision transparency, and (iv) deployment feasibility within healthcare environments.

Structured screening and risk stratification using numerical survey scores and clinical markers are the most task-oriented applications of ML models. Complex representation learning

is typically given less priority in these contexts than consistent performance, repeatability, and interpretable decision methods. More complicated applications, such as multimodal depression severity assessment and longitudinal symptom tracking, are made possible by DL methods, which expand these capabilities by learning hierarchical feature representations directly from raw and multimodal inputs. In contrast, systems that rely on LLMs are best suited for text-intensive tasks that require contextual interpretation and semantic reasoning, such as conversational screening, narrative evaluation, and clinical document analysis.

The ML, DL, and LLM-based methods for depression evaluation can be easily distinguished from one another due to their distinct data needs, interpretability, deployment demands, and bias. For survey-based and resource-constrained clinical investigations, ML approaches work well on small to medium-sized datasets. In contrast, LLMs depend heavily on pre-training and frequently necessitate task-specific customization to conform to clinical goals, and DL methods usually necessitate bigger datasets for strong generalization. The degree of clinical risk is another differentiating factor between these models. High-stakes clinical settings that demand accountability and auditability are better suited to ML models like RF and LR because they provide transparent and traceable decision tracks. In safety-critical operations, DL models may not be suitable due to their poor interpretability, despite their excellent prediction accuracy. Although LLMs are very excellent at language and contextual reasoning, they require constant human supervision due to their opacity, hallucination susceptibility, and prompt sensitivity. Another thing that sets these methods apart is their practical application. In contrast to DL approaches, which frequently necessitate specialized hardware and extensive validation, ML models are very lightweight and can be used in low-resource situations, such as LMICs. Because of the high computational, infrastructure, privacy, and governance requirements, LLMs are not yet applicable to autonomous healthcare contexts.

There is still a significant problem with bias in each of the following three categories: ML and DL techniques largely reflect biases contained in their training datasets, but LLMs may further accentuate broader societal, cultural, and linguistic prejudices acquired from web-scale pre-training. The constraints are particularly important when analyzing depression in under-represented and varied populations.

8.2. Failure Modes and Deployment Risks

Despite their shown efficacy, each methodological paradigm has specific failure modes that prevent their safe application in clinical settings. The generalizability of traditional ML models decreases when trained on small, homogeneous survey populations, when decision thresholds are not properly set, or when features drift. In safety-critical workflows, error identification becomes more challenging when using DL models because to their propensity for overfitting, instability when faced with insufficient data, and poor interpretability. Additional hazards associated with LLM-based systems include hallucinogenic outcomes, responsiveness to quick design, and a lack of foundation in verified clinical knowledge. These constraints highlight the importance of selecting models that are task-aligned and

have ongoing human oversight, especially when making high-risk healthcare decisions.

8.3. Summary Table and Discussion

Table 6 provides a summary of the main evaluation dimensions that differentiate ML, DL, and LLM techniques. Focusing on data requirements, clinical risk, interpretability, and deployment practicality, the review presents a task-oriented and clinically informed comparison methodology for evaluating different techniques in depression evaluation. The comparison shows that methodological applicability is essentially risk-sensitive and task-dependent, reflecting variations in data modality, decision transparency, computing requirements, and real-world implementation limitations. Particularly in settings with limited resources, ML and some DL techniques seem better clinically prepared for organized depression screening and assessment. However, LLMs should not be seen as independent diagnostic systems, but rather as supplementary tools that augment text-intensive analysis and bolster clinician opinion.

Overall, this framework highlights the necessity of aligning model selection with clinical objectives, data availability, acceptable risk thresholds, and practical deployment constraints, rather than relying solely on predictive performance metrics.

9. Challenges in Previous Studies and Future Directions

9.1. Challenges in Previous Studies

9.1.1. Data Limitations and Imbalance

Most of the research on depression analysis that has been done so far relies on biased or small datasets that are derived from certain geographic and language contexts. The lack of established ways for reporting information related to depression, along with privacy concerns and societal stigma, makes data from underdeveloped nations extremely restricted. Also, many of the datasets [114] out there are skewed toward Western populations and their English-language content, which reduces cultural variety and makes it harder to generalize models. Another persistent problem is class imbalance, which results in biased models that are less sensitive to the minority class because non-depressed samples frequently significantly outnumber depressed samples. Due to this disparity, performance in actual clinical situations can be compromised, and evaluation measures can be deemed uncertain. The main difficulties with depression analysis that have been highlighted in earlier research are shown in Figure 15.

9.1.2. Lack of Multimodal Integration

Another repeating problem is the concentration on single-modality data [115]. Many ML studies rely exclusively on survey or questionnaire data (e.g., PHQ-9, DASS-21), whereas most LLM-based studies focus on social media or clinical note text. Depression, however, arises through numerous behavioural and physiological signals, such as speech patterns, facial expressions, sleep quality, and social interaction abnormalities. Without combining different modalities, there is a risk that models will not fully capture the complexity of behaviour that is

associated with depression and associated behaviours. Integrating text, audio, video, and physiological data is a relatively new area of research and has not been adequately studied in order to bridge gaps in relative context and emotional complexity. Even though LLM-based analysis have the capacity to identify depression, they have only been analysed based on textual data, which means that the models have not been validated or tested under multiple cultural, linguistic, or clinical environments [85].

9.1.3. Model Interpretability

Deep learning and LLM-based models frequently function as "black boxes," producing predictions without offering explicit explanations for their results. Clinicians may acknowledge the predictive power of these models, but they are hesitant to put their faith in them when the reasoning behind the prediction is not evident [116]. In the fields of medicine and mental health, knowing the reasoning behind a patient's depression diagnosis is just as crucial as the diagnosis itself. Models that make predictions without providing interpretable reasons are difficult to clinically validate and less suitable for inclusion into established diagnostic procedures.

9.1.4. Ethical and Privacy Concerns

Typically, predicting depression is based on very sensitive data derived from social media, medical records, and chat log information. Although previous research addressed some ethical issues, obtaining consent from users for their data usage, anonymising user data, and preventing potential algorithmic discrimination [117], many ethical considerations were overlooked when accessing such information without the requisite ethical approval. Thus, vulnerability increases among any resulting populations that were already at risk for additional harm. Lastly, if there is a bias present within the training dataset, the resulting trained ML algorithm can potentially continue to promote negative stereotypes about those likewise susceptible individuals based upon gender, age, and ethnicity. Thus, without applying ethical standards and addressing issues of privacy, the successful use of AI and ML within the mental health profession cannot occur.

9.1.5. Insufficient Clinical Validation

The vast majority of current studies on ML and LLM were conducted in either a research or academic environment, using data from retrospective databases. Few have undergone prospective testing or clinical validation with real-world patients [118]. Consequently, model accuracy in controlled datasets may not translate to clinical utility. In addition to data collection method variability and noise, the demographic variability associated with depression contributes to the poor generalizability of the findings associated with most studies. Short-term, cross-sectional study designs limit researchers' ability to gain insight into the long-term progression, relapse patterns, and treatment responses of depression. As a result, ML models may not provide as accurate predictions regarding depression and will require additional longitudinal patient data to be gathered by clinicians.

Table 6: Comparative summary of ML, DL, and LLM approaches for depression analysis across key evaluation dimensions.

Evaluation Dimension	Machine Learning (ML)	Deep Learning (DL)	Large Language Models (LLMs)
Typical Clinical Task	Screening, risk stratification	Severity estimation, multimodal monitoring	Text analysis, conversational screening
Data Modality	Structured numerical data (surveys, clinical variables)	Raw and multimodal data (text, audio, signals)	Unstructured text and conversational data
Sample Efficiency	Effective on small to medium datasets	Requires larger datasets for stable learning	Relies on massive pre-training and adaptation
Computational Cost	Low; suitable for limited hardware	High; often requires GPU acceleration	Very high; typically requires cloud or HPC resources
Interpretability	High; transparent decision logic	Moderate; limited explainability	Low; black-box behaviour
Robustness to Domain Shift	Moderate; sensitive to feature design	Moderate; sensitive to data distribution	Low to moderate; sensitive to language and prompts
Bias and Ethical Risk	Self-report and sampling bias	Dataset and representation bias	Societal, cultural, and language bias amplification
Clinical Deployment Readiness	High; widely applicable	Moderate; requires validation	Low; requires strict human oversight

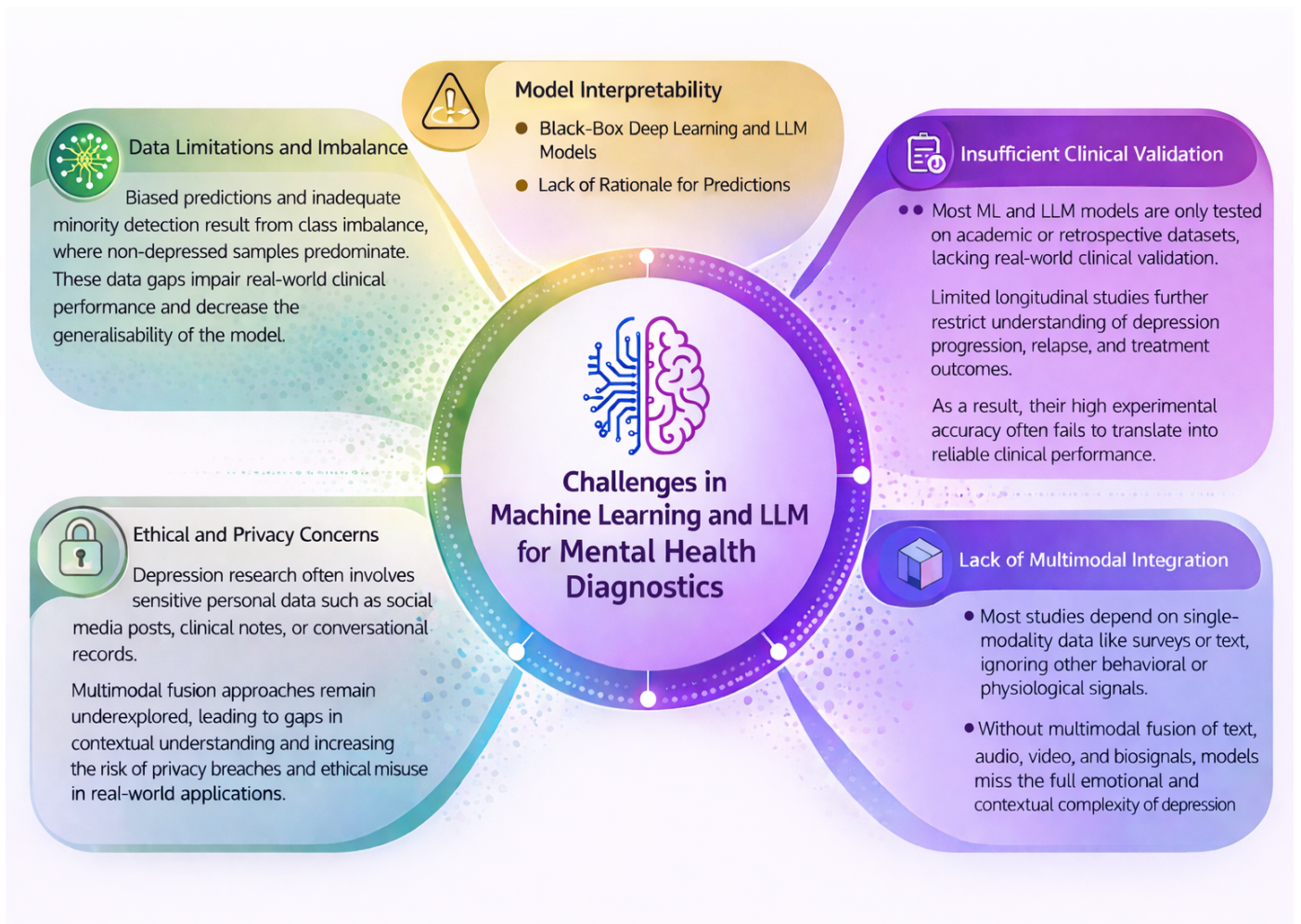


Figure 15: Challenges in ML and LLM for mental health diagnosis.

9.2. Future Research Directions in Depression Analysis

Research in artificial intelligence (AI) for depression analysis should progress away from proof-of-concept studies and toward frameworks that are clinically grounded, methodologically rigorous, and deployment-oriented. While ML and LLM-based techniques have shown promise in previous research, problems with dataset diversity, external validation, interpretability, privacy concerns, and clinical trials have hampered their practical application. Therefore, the following priority directions should be the center of attention for future work.

9.2.1. Construction of Large-Scale, Multilingual, and Clinically Representative Datasets

The development of extensive datasets that more accurately represent the clinical, linguistic, cultural, and demographic diversity of depression is a top research aim. The majority of current models have limited generalizability across populations and healthcare facilities since they were created using tightly specified cohorts, such as student surveys, social media users, or single-site retrospective samples. Thus, future datasets should incorporate behavioral data from both clinical and community contexts, longitudinal symptom records, multilingual clinical narratives, and sociodemographic data. To lessen regional and cultural bias, extra emphasis should be paid to underrepresented areas, including South Asia, Africa, and other low-resource environments. In order to determine whether LLM-generated samples increase robustness without adding artificial linguistic artifacts or exacerbating hidden biases, future research should also compare actual and synthetic data augmentation techniques.

9.2.2. Task-Specific Modeling Rather Than General Depression Classification

Depression analysis should not be treated as a generic categorization task in future investigations. Screening at an early stage, estimating severity, predicting relapse, treatment-response forecasting, monitoring risk escalation, and tracking symptoms over time are all clinically important aims with their own labels, thresholds, and outcomes. Consequently, rather than depending just on overall binary classification performance, research ought to center over task-specific models and evaluation methodologies. In early screening, sensitivity and calibration should be top priorities, whereas in relapse prediction, temporal resilience and false negative control should be front and center. The therapeutic relevance of depression analysis may be substantially enhanced if it was reframed through subtasks with clear clinical definitions.

9.2.3. Multimodal Fusion for Robust and Context-Aware Depression Assessment

Depressive symptoms can be picked up in text, but they can also be seen in speech, facial expressions, sleep patterns, social behavior, and physical signals. Subsequent research should thus emphasize multimodal frameworks that cohesively combine textual, audio, visual, physiological, and behavioral data over time. Rather of relying on basic feature concatenation, it is advisable to investigate more sophisticated methodologies,

such as cross-modal attention, temporal fusion transformers, graph-based fusion models, and uncertainty-aware architectures, especially for addressing missing or noisy inputs. In an effort to ascertain whether multimodal systems can surpass text-only models in the areas of early detection, relapse prediction, and treatment monitoring, longitudinal studies that integrate ubiquitous sensors, smartphone usage patterns, and conversational data are particularly necessary.

9.2.4. Hybrid ML–LLM Architectures for Structured and Unstructured Clinical Evidence

A promising research avenue is the creation of hybrid pipelines that integrate traditional ML with LLM-based systems. In these architectures, ML models can process structured data, including psychometric scores, demographic variables, comorbidities, medication history, and behavioral indicators, whereas LLMs analyze unstructured sources such as clinical notes, patient narratives, therapy transcripts, and social media text. Future research should investigate whether fusion strategies, such as feature-level fusion, late fusion, or uncertainty-weighted ensembles, enhance predictive consistency, calibration, and interpretability relative to independent models. Special attention must be directed towards retrieval-grounded LLMs that utilize reliable clinical evidence to mitigate hallucination and enhance traceability. Comparative research must assess the trade-offs between full fine-tuning, parameter-efficient tuning, and smaller domain-adapted models, particularly in low-resource deployment contexts.

9.2.5. Explainable and Ethical Model Design

Explainability should be a central requirement in future depression analysis systems. Studies should quantitatively assess the contribution of behavioural, linguistic, and physiological features using established XAI techniques such as SHAP, LIME, and attention visualisation. Future work must also evaluate ethical risks, including bias propagation and privacy leakage, while ensuring compliance with clinical data protection standards such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA).

9.2.6. Failure Mode Analysis and Risk-Aware Decision Support

Future systems should define failure modes instead of providing average performance. Language, sarcasm, domain shift, class imbalance, missing modalities, prompt instability, and complicating social factors can cause depression analysis errors. The next study should identify these failure scenarios and develop risk-aware measures including confidence-based triage, human escalation, conservative thresholding, and safety-oriented abstention. This involves testing LLM-based systems for prompt sensitivity, hallucination, and instability across rephrasings and ML-based systems for dataset shift and incomplete records. Instead of an autonomous diagnostic tool, AI should be a supervised clinical decision-support system with specified operating restrictions.

9.2.7. Prospective Clinical Validation and Real-World Deployment Studies

A significant deficiency in the research is the absence of prospective and workflow-level validation. Most studies use retrospective datasets and controlled experimental conditions, making it uncertain if their success will translate into therapeutic benefit. Future studies should evaluate depression evaluation systems in real-world settings via silent deployment, clinician-in-the-loop investigations, pragmatic trials, and longitudinal monitoring. These evaluations should include referral quality, alert burden, clinician acceptance, patient safety, override rates, and follow-up results, in addition to discriminatory measures. Even extremely accurate models are scientifically promising but clinically untested without prospective validation.

9.2.8. Privacy-Preserving and Federated Learning for Sensitive Mental Health Data

Depression analysis commonly uses sensitive personal data; thus, future research must focus privacy-preserving training and deployment. Federated learning, differential privacy, secure aggregation, split learning, and on-device inference should be prioritized in hospital networks and mobile mental health platforms where centralized data sharing may be ethically or legally prohibited. Beyond suggesting these strategies, future studies should assess their implications on model performance, fairness, and calibration. This research is crucial to establishing scalable depression assessment systems that meet regulatory and public expectations for confidentiality and data governance.

In general, clinically validated, multimodal, interpretable, privacy-preserving, and fairness-aware artificial intelligence systems that can consistently function across varied populations and real-world care settings should be the primary focus of the next phase of depression analysis research. For ML and LLM-based techniques to become reliable tools for depression evaluation and mental health decision support, there must be a shift from retrospective testing to robust clinical implementation.

10. Conclusion

This review indicates that artificial intelligence is increasingly serving as a significant facilitator of depression evaluation; nonetheless, its impact is primarily contingent upon specific tasks rather than universally applicable models. Because of its better interpretability, reduced computing load, and increased deployment availability in resource-constrained contexts, traditional ML is still the best practical choice for structured clinical and psychometric variables, according to the literature. This capability can be expanded by DL to include more complicated and multimodal representations, and LLMs are particularly useful for text-intensive tasks that demand contextual and semantic reasoning, such as clinical document analysis, narrative assessment, and conversational screening. As a result, the CRRCF is the main takeaway from this review. It rethinks comparison by adding clinically relevant dimensions like task fit, data modality, interpretability, deployment feasibility, and patient-safety risk to the mix, rather than focusing

solely on isolated performance metrics. The review suggests that the field is yet methodologically underdeveloped for regular autonomous clinical implementation. A significant portion of the published research originates from internally vetted, retrospective, student-centered, or otherwise restricted datasets, which constrains generalizability and exaggerates real-world applicability. Documented performance improvements are frequently influenced by preprocessing, feature engineering, and dataset construction, in addition to model architecture, while ongoing issues related to bias, privacy, limited prospective validation, and awareness of failure modes remain unaddressed. In LLM-based systems, quick sensitivity, the risk of hallucination, and opacity exacerbate these issues. Future advancements should thus include clinically representative multilingual datasets, task-specific evaluations, multimodal and hybrid ML–LLM pipelines, enhanced explainability, and projected human-supervised validation. The domain should eventually be evaluated not solely on benchmark accuracy, but on its capacity to provide dependable, ethical, and contextually relevant assistance for depression screening and clinical decision-making.

Data availability

No data was used for the research described in the article.

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